A MICROCANONICAL PHASE TRANSITION FOR THE COLLATZ AFFINE RANDOM MODEL

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Abstract. Tao's proof that almost all Collatz orbits attain almost bounded values proceeds through a Syracuse affine random model whose n-step map is $\mathrm{Aff}_{a_1,\dots,a_n}(x) = 3^n 2^{-S_n} x + F_n(a_1,\dots,a_n)$ with a_i i.i.d. geometric variables of mean 2. Tao establishes superpolynomial 3-adic Fourier decay of the offset F_n and remarks (Remark 1.16) that an upgrade from logarithmic density to natural density would likely require fine-scale mixing of the entire affine map.

We isolate the multiplicative marginal $2^{-S_n} \pmod{3^k}$ and prove a sharp threshold: it equidistributes precisely up to $3^k \ll \sqrt{n}$, with a complete Jacobi-theta cutoff profile in the critical window $M_k/\sqrt{n} \to \lambda$. Beyond this scale the marginal is asymptotically maximally far from uniform, so any extension of Tao's mixing to refined moduli must exploit the coupling between S_n and F_n.

We then introduce a microcanonical framework: condition on $S_n = s$. Under this conditioning the multiplicative phase becomes deterministic, and the affine Fourier coefficient at any frequency $\xi \in \mathbb{Z}/3^{n+k}\mathbb{Z}$ with $v_3(\xi) < k$ is reduced to a primitive ternary Bernoulli-bridge transform on a real-line perpetuity $\mathcal{F}_\infty$. The law of the perpetuity is the stationary measure of a contracting-on-average two-map affine system whose linear ratios are $1/2$ and $3/2$, which are not integral powers of a common λ with $1/\lambda$ Pisot; Brémont's classification therefore yields $\widehat{\mu}_{\mathcal{F}}(t) \to 0$ as $|t| \to \infty$.

Combining ensemble equivalence, an exact suffix self-similarity, and Brémont's Rajchman theorem we prove a sharp phase transition for the worst (resonant) hard frequency $\eta_s = 2^s$:

$$\Delta_n := s \log_3 2 - n - h \to +\infty \implies \text{microcanonical coefficient} \to 0,$$

$$\Delta_n \to -\infty \implies \text{microcanonical coefficient} \to 1.$$

The transition occurs precisely at the entropy line $h = h_* n + o(n)$ with $h_* = 2\log_3 2 - 1 \approx 0.262$. We further reduce the full hard-range mixing problem to a single *primitive ternary bridge decay* statement and prove the $o(m/\log m)$-frequency case unconditionally. The remaining range is identified as a clean one-dimensional Diophantine question.

1. Introduction

1.1. **Background.** Write $\mathbb{N}+1 := \{1,2,3,\dots\}$ for the positive integers. The Collatz map $\mathrm{Col}\colon \mathbb{N}+1 \to \mathbb{N}+1$ is defined by $\mathrm{Col}(N) = 3N+1$ for N odd and $\mathrm{Col}(N) = N/2$ for N even; we refer to [4] for a comprehensive survey. Previously, Korec [3] showed that $\mathrm{Col}_{\min}(N) \le N^\theta$ for almost all N (natural density) whenever $\theta > \log 3/\log 4$. Tao [11] proved that for every $f\colon \mathbb{N}+1 \to \mathbb{R}$ with $f(N) \to \infty$, $\mathrm{Col}_{\min}(N) := \inf_{m\ge 0} \mathrm{Col}^m(N) \le f(N)$ for almost all N in the sense of *logarithmic* density. The proof passes to the Syracuse map and to an affine random model: with $a_1,\dots,a_n$ i.i.d. copies of $\mathrm{Geom}(2)$, where $\mathbb{P}(\mathrm{Geom}(2) = m) = 2^{-m}$ for $m \ge 1$, the n-step affine map is

$$\mathrm{Aff}_{a_1,\dots,a_n}(x) = 3^n 2^{-S_n} x + F_n(a_1,\dots,a_n), \qquad S_n := a_1 + \cdots + a_n, \tag{1}$$

with the Syracuse offset

$$F_n(a_1,\dots,a_n) := \sum_{m=1}^{n} 3^{n-m} 2^{-a_{[m,n]}}, \qquad a_{[m,n]} := a_m + \cdots + a_n. \tag{2}$$

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Tao's key analytic input is a fine-scale 3-adic mixing statement for the offset F_n, equivalent (via Plancherel) to a superpolynomial Fourier-decay estimate

$$\left| \mathbb{E}\, e^{-2\pi i \xi \mathrm{Syrac}(\mathbb{Z}/3^n\mathbb{Z})/3^n} \right| \ll_A n^{-A} \tag{3}$$

uniformly in n and primitive $\xi \in \mathbb{Z}/3^n\mathbb{Z}$ [11, Prop. 1.17], where $\mathrm{Syrac}(\mathbb{Z}/3^n\mathbb{Z}) \equiv F_n \pmod{3^n}$. Tao remarks that an upgrade to natural density appears to require a corresponding fine-scale mixing statement for the *entire* random affine map, not just for F_n [11, Rem. 1.16].

1.2. **Where the multiplicative marginal fails.** A direct attack on Tao's Remark 1.16 requires understanding the multiplicative marginal $2^{-S_n} \pmod{3^k}$ in addition to the offset. Our starting point, established in earlier work [10], is that this marginal mixes only up to a logarithmic scale. For each $k \geq 1$ let $G_k := (\mathbb{Z}/3^k\mathbb{Z})^\times$ and $M_k := \#G_k = 2 \cdot 3^{k-1}$. Define the additive walk $Y_{n,k} := -S_n \pmod{M_k}$, identified with the multiplicative process via the discrete logarithm base 2, and write $\nu_{n,k} := \mathcal{L}(Y_{n,k})$, U_{M_k} for the uniform law.

Theorem 1.1 ([10]; see Theorem 2.1 for an independent proof with cutoff profile)**.** *For any $k = k(n)$:*

(a) *if $M_k = o(\sqrt{n})$, equivalently $k \leq \frac{1}{2}\log_3 n - \omega(1)$, then $\|\nu_{n,k} - U_{M_k}\|_{\mathrm{TV}} \to 0$;*
(b) *if $\sqrt{n} = o(M_k)$, equivalently $k \geq \frac{1}{2}\log_3 n + \omega(1)$, then $\|\nu_{n,k} - U_{M_k}\|_{\mathrm{TV}} \to 1$.*

The present paper extends [10] in several directions. First, we close the critical window of Theorem 1.1 with a complete cutoff profile. Second, we develop a microcanonical (conditional on $S_n = s$) Fourier analysis of the affine map and obtain a sharp phase transition for the worst hard frequencies at the entropy line. Third, we reduce the full hard-frequency problem to an explicit primitive ternary Bernoulli-bridge transform, and prove its $o(m/\log m)$-frequency case.

1.3. **The microcanonical viewpoint.** Throughout the paper we write $\mathbf{e}_Q(x) := e^{2\pi i x/Q}$ and use $Q = 3^N$ with various N. Splitting the affine character at $Q = 3^{n+k}$ gives

$$\mathbf{e}_Q\big(\xi \mathrm{Aff}_{a_1,\ldots,a_n}(x)\big) = \mathbf{e}_{3^k}\big(\xi 2^{-S_n} x\big)\, \mathbf{e}_{3^{n+k}}\big(\xi F_n\big). \tag{4}$$

Theorem 1.1 kills any naive attempt to mix the multiplicative factor 2^{-S_n} for $3^k \gg \sqrt{n}$. However, conditioning on $S_n = s$ turns the multiplicative factor into a deterministic phase $\mathbf{e}_{3^k}(\xi 2^{-s} x)$; only the offset Fourier coefficient $\mathbb{E}[\mathbf{e}_{3^{n+k}}(\xi F_n) \mid S_n = s]$ remains stochastic. This is the principle that lets the analysis bypass the multiplicative obstruction.

1.4. **Main results.** We classify frequencies $\xi \in \mathbb{Z}/3^{n+k}\mathbb{Z}$ by their *effective ternary depth* $d(\xi) := n + k - v_3(\xi)$, splitting into easy, subdiffusive and hard regimes. We write $h := k - v_3(\xi)$, so $d(\xi) = n + h$. Frequencies with $h = 0$ reduce to Tao's setting, while frequencies with $h \geq 1$ ($v_3(\xi) < k$) are the new "hard" frequencies that probe the finer modulus 3^{n+k}.

For central bridges $S_n = s$ with $s = 2n + O(\sqrt{n \log n})$ define

$$\Delta_n := s \log_3 2 - n - h. \tag{5}$$

The number $h_* := 2\log_3 2 - 1 \approx 0.2618595$ is the entropy boundary: $\Delta_n > 0$ if and only if $h < h_* n + O(\sqrt{n \log n})$. The constant h_* arises from the Shannon entropy $H(\mathrm{Geom}(2)) = 2\log 2$ versus the modulus entropy $(n + h)\log 3$.

Theorem 1.2 (Sharp resonant phase transition; Theorem 8.6)**.** *Let $s = 2n + O(\sqrt{n \log n})$, $h \geq 1$, and let $\xi = 3^{k-h} \cdot 2^s$ with $h \leq k$ be the resonant hard frequency.*

(1) (Subcritical decay). *If $\Delta_n \to +\infty$, then*

$$\mathbb{E}\big[\, \mathbf{e}_{3^{n+k}}(\xi F_n) \;\big|\; S_n = s\big] \to 0.$$

(2) (Supercritical obstruction). *If $\Delta_n \to -\infty$, then*

$$\big|\mathbb{E}[\mathbf{e}_{3^{n+k}}(\xi F_n) \mid S_n = s]\big| \to 1.$$

(3) (Critical profile). *If* $3^{\Delta_n} \to \lambda \in (0,\infty)$,

$$\mathbb{E}\big[\mathbf{e}_{3^{n+k}}(\xi F_n) \bigm| S_n = s\big] - \mathbf{e}_{3^{n+h}}(3^{n-1})\, \widehat{\mu}_{\mathcal{F}}(\lambda) \to 0,$$

where $h = k - v_3(\xi)$. *The deterministic phase equals* $\exp(2\pi i/3^{h+1})$, *which tends to* 1 *when* $h \to \infty$ *(the entropy-critical regime, since* $3^{\Delta_n} \to \lambda \in (0,\infty)$ *forces* $h \sim h_* n$*). The perpetuity* $\mathcal{F}_\infty$ *satisfies* $\mathcal{F}_\infty \overset{d}{=} 2^{-A}(1 + 3\mathcal{F}'_\infty)$ *with* $A \sim \mathrm{Geom}(2)$ *and* $\mathcal{F}'_\infty$ *an independent copy.*

The same conclusions hold for the full affine character (4), *since under* $S_n = s$ *the multiplicative factor is a deterministic phase.*

The proof has three main ingredients. First, an exact algebraic identity: multiplying F_n by 2^s converts the 3-adic offset phase into a real-line Bernoulli-bridge integral

$$\mathbb{E}[\mathbf{e}_{3^N}(2^s F_n) \mid S_n = s] = \mathbf{e}_{3^N}(3^{n-1}) \cdot \mathbb{E}_{\Omega_{m,r}} \exp(2\pi i \lambda_n F_n^{\mathbb{R}}), \quad \lambda_n := \frac{2^s}{3^N} = 3^{\Delta_n}, \tag{6}$$

where $F_n^{\mathbb{R}} = \sum_r 3^r\, 2^{-(A_0+\cdots+A_r)}$ is the offset viewed as a real number rather than a 3-adic residue (Section 6). This is the essential observation that converts the multiplicative obstruction to a real-line Diophantine question.

Second, the limiting object as $n \to \infty$ is the perpetuity $\mathcal{F}_\infty := \sum_{r\geq 0} 3^r\, 2^{-(A_0+\cdots+A_r)}$, the stationary measure of the contracting-on-average two-map IFS $\{\psi_0(x) = x/2,\ \psi_1(x) = 1/2 + (3/2)x\}$. The linear ratios $\{1/2, 3/2\}$ are not powers of a common Pisot number, so by Brémont's classification [1, Theorem 2.5], $\widehat{\mu}_{\mathcal{F}}(t) \to 0$ as $|t| \to \infty$ (Section 7).

Third, an exact suffix self-similarity: for any decomposition $m = J + L$,

$$X_m = X_J + 3^{r-y} \cdot 2^{-J} \cdot X_L(\omega), \tag{7}$$

which collapses a global bridge into a smaller bridge of length L with shifted parameters. This bootstraps the near-critical decay $\Delta_n = o(n/\log n)$ (Theorem 8.3) to the entire subcritical range $\Delta_n \to +\infty$ via an inductive descent on the suffix length (Section 8); the residual range $\Delta_n \geq n/\log n$ is reduced by suffix self-similarity to a smaller bridge whose entropy gap is $o(L/\log L)$, so that Theorem 8.3 applies at the smaller scale.

Theorem 1.3 (Hard-range decay at logarithmic oversampling; Theorem 5.1)**.** *Fix* $K, L, A > 0$. *Suppose* $|s - 2n| \leq L\sqrt{n \log n}$. *Then*

$$\sup_{\substack{\xi \in \mathbb{Z}/3^{n+k}\mathbb{Z} \\ 1 \leq k - v_3(\xi) \leq K \log n}} \big|\mathbb{E}[\mathbf{e}_{3^{n+k}}(\xi F_n) \mid S_n = s]\big| \ll_{A,K,L} n^{-A}.$$

This is proved by Tao's renewal estimate combined with an augmentation trick (Section 5). Beyond logarithmic oversampling we obtain a complete reduction:

Theorem 1.4 (Reduction to a Bernoulli-bridge transform; Theorem 6.2)**.** *For every primitive* $\eta \in (\mathbb{Z}/3^{n+h}\mathbb{Z})^\times$ *and any* $s \geq n$, *let* $q \in \mathbb{Z}$ *be the unique representative of* $\eta \cdot 2^{-s} \pmod{3^{n+h}}$ *in* $(-3^{n+h}/2, 3^{n+h}/2]$. *Then* $3 \nmid q$ *and*

$$\mathbb{E}[\mathbf{e}_{3^{n+h}}(\eta F_n) \mid S_n = s] = \mathbf{e}_{3^{n+h}}(q \cdot 3^{n-1}) \cdot \mathbb{E}_{\Omega_{m,r}} \mathbf{e}_{3^{n+h}}(2qP_m),$$

where $m = s - 1$, $r = n - 1$, $\Omega_{m,r} = \{\varepsilon \in \{0,1\}^m : \sum \varepsilon_j = r\}$, $X_m := \sum_{j=1}^m \varepsilon_j\, 3^{R_{j-1}}\, 2^{-j}$, $R_j := \varepsilon_1 + \cdots + \varepsilon_j$, *and* $P_m := 2^m X_m \in \mathbb{Z}$.

Finally, we prove a subexponential decay theorem for the bridge transform that already covers a non-trivial portion of the hard range:

Theorem 1.5 (Subexponential primitive-frequency decay; Theorem 9.2)**.** *Let* $r = m/2 + O(\sqrt{m \log m})$ *and let* $t_m = q_m\, 2^{m+1}/3^{\mathfrak{N}_m}$ *with* $3 \nmid q_m$. *If* $|t_m| \to \infty$ *and* $\log|t_m| = o(m/\log m)$, *then*

$$\mathbb{E}_{\Omega_{m,r}} \mathbf{e}_{3^{\mathfrak{N}_m}}(2q_m P_m) \to 0.$$

For the exponential range $\log|q| = \Theta(m)$ we prove a strong density-one form of HFBD.

Theorem 1.6 (Density-one and fiberwise flattening; Theorems 10.3, 10.4)**.** *Let $L = 2P$, $y = P + O(\sqrt{P \log P})$, and $d > y$. Then there is $c > 0$ such that*

$$\frac{1}{\varphi(3^d)} \sum_{\substack{q \bmod 3^d \\ 3 \nmid q}} |\mathcal{C}_{L,y,d}(q)|^2 \ll L^C \, e^{-cL},$$

and for every $0 \le d_0 < d$ and every primitive $q_0 \in (\mathbb{Z}/3^{d_0}\mathbb{Z})^\times$,

$$\frac{1}{\#\{q \equiv q_0 \pmod{3^{d_0}}\}} \sum_{\substack{q \bmod 3^d \\ q \equiv q_0 \pmod{3^{d_0}}}} |\mathcal{C}_{L,y,d}(q)|^2 \ll L^C \, e^{-c\min(d-d_0,L)}.$$

Iterating across a depth chain $0 = d_0 < d_1 < \cdots < d_T$ produces a multiscale exceptional set $\mathcal{E}_T \subset (\mathbb{Z}/3^{d_T}\mathbb{Z})^\times$ of positive 3-adic codimension (Corollary 10.5).

The density-one half of Theorem 1.6 is proved by an explicit pair-switch averaging that transfers 1/2 contraction per "active" pair and is closed by a saddle-point estimate on the bridge generating function $(1 + z + z^2)^P/(1 + 2z + z^2)^P$. The fiberwise version restricts the averaging to a 3^{d_0}-cylinder and exploits that "active pairs early in the bridge" contribute fresh ternary digits beyond depth d_0. We state the supremum form as a conjecture and discuss the residual rigidity obstruction.

1.5. **Relation to Tao's Remark 1.16.** Theorem 1.2 resolves the worst (resonant) hard frequency in Tao's Remark 1.16 throughout the entire entropy-allowed range: the conditional affine Fourier coefficient decays for $h < h_* n - \omega(1)$ and fails to decay for $h > h_* n + \omega(1)$. Theorem 1.3 extends this to all hard frequencies in the logarithmic-oversampling regime, and Theorem 1.4 clarifies what an unconditional natural-density upgrade would require beyond what Tao's machinery already supplies. The matching upper and lower bounds at the entropy line $h = h_* n$ identify a hard ceiling: above $h_* n$, the white-point method cannot yield uniform mixing, regardless of any technical strengthening of Tao's renewal estimate.

1.6. **Notation and conventions.** We write $f \ll g$ for $f = O(g)$ and $f \ll_A g$ when the implied constant depends on parameter A. $\omega(1)$ denotes any sequence tending to ∞. We use $\mathbf{e}_Q(x) = e^{2\pi i x/Q}$ and reserve $Q = 3^N$ for various N. Probabilities and expectations are taken under $(a_1, \ldots, a_n) \sim \mathrm{Geom}(2)^{\otimes n}$ unless otherwise specified; *microcanonical* expectations $\mathbb{E}[\,\cdot \mid S_n = s]$ condition on the total $S_n = s$.

1.7. **Organization.** Section 2 proves the cutoff profile of Theorem 1.1. Section 3 sets up the effective-depth stratification and proves a fixed-depth limit identifying the obstruction at small depths. Section 4 proves subdiffusive microcanonical decay. Section 5 proves Theorem 1.3. Section 6 establishes the Bernoulli-bridge normal form and the resonant perpetuity. Section 7 verifies that Brémont's classification gives the Rajchman property of the perpetuity. Section 8 proves the resonant phase transition (Theorem 1.2) via suffix self-similarity. Section 9 applies the bridge reduction to prove the $o(m/\log m)$-frequency decay Theorem 1.5. Section 10 proves the density-one and fiberwise flattening estimates (Theorems 10.3, 10.4) and derives the multiscale codimension bound for the exceptional tree (Corollary 10.5). Section 11 states Conjecture 11.1 (HFBD-sup) and discusses the residual rigidity question. Appendix A collects auxiliary technical lemmas.

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2. Cutoff profile for the multiplicative marginal

This section refines Theorem 1.1 by computing the limiting total-variation profile in the critical window $M_k/\sqrt{n} \to \lambda$.

For a modulus $M \geq 1$ let μ_M be the law of $-\mathrm{Geom}(2)$ on $\mathbb{Z}/M\mathbb{Z}$:

$$\mu_M(r) = \mathbb{P}(-\mathrm{Geom}(2) \equiv r \pmod{M}), \qquad r \in \mathbb{Z}/M\mathbb{Z}. \tag{8}$$

The Fourier transform $\widehat{\mu}_M(j) = \sum_r \mu_M(r) e^{2\pi i j r/M}$ admits the closed form

$$\widehat{\mu}_M(j) = \frac{\frac{1}{2}e^{-2\pi i j/M}}{1 - \frac{1}{2}e^{-2\pi i j/M}}, \qquad |\widehat{\mu}_M(j)|^2 = \frac{1}{1 + 8\sin^2(\pi j/M)}, \tag{9}$$

established by summing the geometric series; see [10, §3]. The Fourier transform of $\nu_{n,k} = \mathcal{L}(-S_n \bmod M_k)$ at j is $\widehat{\nu}_{n,k}(j) = \widehat{\mu}_{M_k}(j)^n$.

Theorem 2.1 (Cutoff profile)**.** *Suppose $M_k/\sqrt{n} \to \lambda \in (0,\infty)$ as $n \to \infty$. Define*

$$H_\lambda(t) := \sum_{j\in\mathbb{Z}} e^{-4\pi^2 j^2/\lambda^2} e^{-2\pi i j t} = \frac{\lambda}{\sqrt{4\pi}} \sum_{\ell\in\mathbb{Z}} \exp\left(-\frac{\lambda^2(t+\ell)^2}{4}\right), \qquad t \in \mathbb{R}/\mathbb{Z}, \tag{10}$$

where the second equality follows from Poisson summation. Then

$$\sup_{x\in\mathbb{Z}/M_k\mathbb{Z}} \left| M_k\, \nu_{n,k}(x) - H_\lambda\left(\frac{x+2n}{M_k}\right)\right| \to 0, \tag{11}$$

and

$$\|\nu_{n,k} - U_{M_k}\|_{\mathrm{TV}} \to \frac{1}{2}\int_0^1 |H_\lambda(t) - 1|\, dt. \tag{12}$$

Proof. We have $\mathbb{E}[a_1] = 2$ and $\mathrm{Var}(a_1) = 2$ since $a_1 \sim \mathrm{Geom}(2)$ on $\{1, 2, \dots\}$ has $\mathbb{E}[a_1] = \sum_{m\geq 1} m2^{-m} = 2$ and $\mathbb{E}[a_1^2] = \sum_{m\geq 1} m^2 2^{-m} = 6$. For fixed $j \in \mathbb{Z}$ and $M = M_k \sim \lambda\sqrt{n}$, the cumulant expansion at $\theta = 2\pi j/M$ gives

$$\begin{aligned}\log \widehat{\mu}_M(j) &= -i\theta\, \mathbb{E}[a_1] - \frac{\theta^2}{2}\mathrm{Var}(a_1) + O_j(M^{-3})\\ &= -\frac{4\pi i j}{M} - \frac{4\pi^2 j^2}{M^2} + O_j(M^{-3}),\end{aligned}$$

where the implied constant depends on j. For each fixed j, since $n/M^2 \to 1/\lambda^2$,

$$\widehat{\mu}_M(j)^n \to \exp\left(-\frac{4\pi i j n}{M}\right)\exp\left(-\frac{4\pi^2 j^2}{\lambda^2}\right).$$

Fourier inversion and frequency split. By Fourier inversion on $\mathbb{Z}/M\mathbb{Z}$,

$$M\, \nu_{n,k}(x) = \sum_{j\in\mathbb{Z}/M\mathbb{Z}} \widehat{\mu}_M(j)^n\, e^{-2\pi i j x/M}. \tag{13}$$

Uniform Gaussian tail. For $0 \leq j \leq M/2$, $\sin u \geq 2u/\pi$ on $[0, \pi/2]$ together with (9) gives $|\widehat{\mu}_M(j)|^2 \leq 1/(1 + 32(j/M)^2)$, hence $|\widehat{\mu}_M(j)|^n \leq \exp\bigl(-c\,n(j/M)^2\bigr)$ for some absolute $c > 0$. With $n/M^2 \to 1/\lambda^2$, for all sufficiently large n,

$$|\widehat{\mu}_M(j)|^n \leq \exp(-c_\lambda j^2) \qquad (0 \leq j \leq M/2),$$

with $c_\lambda > 0$. By symmetry $j \leftrightarrow M - j$, the same bound holds for $j \in [M/2, M]$ upon writing $\|j\|_M := \min(j, M-j)$.

Pointwise convergence at fixed cutoff. Fix $J \geq 1$. For $|j| \leq J$, the cumulant expansion gives $\widehat{\mu}_M(j)^n \to e^{-4\pi i j n/M - 4\pi^2 j^2/\lambda^2}$ pointwise. The partial sum

$$\sum_{\|j\|_M \leq J} \widehat{\mu}_M(j)^n\, e^{-2\pi i j x/M} \to \sum_{|j|\leq J} e^{-4\pi^2 j^2/\lambda^2} e^{-2\pi i j(x+2n)/M}$$

uniformly in $x \in \mathbb{Z}/M\mathbb{Z}$ as $n \to \infty$.

Tail $J\to\infty$. The remainder satisfies, uniformly in x and n sufficiently large,

$$\Big|\sum_{\|j\|_M>J}\widehat{\mu}_M(j)^n\,e^{-2\pi ijx/M}\Big|\le 2\sum_{j>J}e^{-c_\lambda j^2}\to 0\quad\text{as } J\to\infty.$$

Setting $t:=(x+2n)/M \bmod 1$ and combining,

$$M\,\nu_{n,k}(x)\to\sum_{j\in\mathbb{Z}}e^{-4\pi^2j^2/\lambda^2}e^{-2\pi ijt}=H_\lambda(t)\tag{14}$$

uniformly in $x\in\mathbb{Z}/M_k\mathbb{Z}$.

Poisson summation. The Gaussian $g(u):=(\lambda/\sqrt{4\pi})\,e^{-\lambda^2u^2/4}$ has Fourier transform (with the convention $\widehat{g}(\xi)=\int g(u)e^{-2\pi i\xi u}\,du$) $\widehat{g}(\xi)=e^{-4\pi^2\xi^2/\lambda^2}$. Poisson summation $\sum_\ell g(t+\ell)=\sum_j\widehat{g}(j)e^{2\pi ijt}$ yields the closed form in (10).

Total variation. The total-variation identity is $\|\nu_{n,k}-U_{M_k}\|_{\mathrm{TV}}=\frac12M_k^{-1}\sum_x|M_k\,\nu_{n,k}(x)-1|$. By (14) and uniform convergence, the right-hand side is a Riemann sum of $\frac12|H_\lambda(t)-1|$ on $[0,1]$ with mesh $1/M_k\to0$, which converges to $\frac12\int_0^1|H_\lambda(t)-1|\,dt$. $\square$

3. Effective ternary depth and a fixed-depth obstruction

We now introduce the basic stratification of frequencies that drives the remainder of the paper. For $\xi\in\mathbb{Z}/Q\mathbb{Z}$ with $Q=3^{n+k}$ and $\xi\ne0$, let $v=v_3(\xi)$ be its 3-adic valuation and set $d=d(\xi)=n+k-v$, so $\xi=3^v\eta$ with $\eta\in(\mathbb{Z}/3^d\mathbb{Z})^\times$. Then $\mathbf{e}_Q(\xi F_n)=\mathbf{e}_{3^d}(\eta F_n)$.

Lemma 3.1 (Tail reduction). *For every $1\le d\le n$ and every $(a_1,\dots,a_n)$,*

$$F_n(a_1,\dots,a_n)\equiv F_d(a_{n-d+1},\dots,a_n)\pmod{3^d},$$

where F_d is the Syracuse offset (2) *on the last d variables.*

Proof. Inspect (2) term by term. For $m\le n-d$ one has $n-m\ge d$, so the term $3^{n-m}2^{-a_{[m,n]}}$ is divisible by 3^d and vanishes modulo 3^d. Setting $r=m-(n-d)$ in the remaining range $m=n-d+1,\dots,n$ gives $3^{n-m}=3^{d-r}$ and $a_{[m,n]}=a_{n-d+r}+\cdots+a_n$, recovering $F_d(a_{n-d+1},\dots,a_n)$ termwise. $\square$

A naive hope would be that all non-trivial frequencies decay under microcanonical conditioning. The next result rules this out at fixed effective depth.

Proposition 3.2 (Fixed-depth limit). *Let $d\ge1$ and $\eta\in(\mathbb{Z}/3^d\mathbb{Z})^\times$ be fixed. For any $s=s_n$ with $s/n\to2$,*

$$\mathbb{E}\big[\mathbf{e}_{3^d}(\eta F_n)\;\big|\;S_n=s\big]\to\mathbb{E}\big[\mathbf{e}_{3^d}(\eta F_d(A_1,\dots,A_d))\big],$$

where $(A_1,\dots,A_d)$ are i.i.d. Geom(2). *For $d=1$ and $\eta=1$ the limit equals $\frac13e^{2\pi i/3}+\frac23e^{4\pi i/3}$, of modulus $1/\sqrt3$.*

Proof. By Lemma 3.1, $\mathbf{e}_{3^d}(\eta F_n)=\mathbf{e}_{3^d}(\eta F_d(a_{n-d+1},\dots,a_n))$. We claim that the conditional law $\mathcal{L}((a_{n-d+1},\dots,a_n)\mid S_n=s)$ converges in total variation to $\mathrm{Geom}(2)^{\otimes d}$ as $n\to\infty$ with $s/n\to2$.

Indeed, for fixed $b=(b_1,\dots,b_d)$ with $B:=b_1+\cdots+b_d$,

$$\begin{aligned}\mathbb{P}\big(a_{n-d+1}=b_1,\dots,a_n=b_d\;\big|\;S_n=s\big)&=\frac{2^{-B}\,\mathbb{P}(S_{n-d}=s-B)}{\mathbb{P}(S_n=s)}\\&=\frac{\binom{s-B-1}{n-d-1}}{\binom{s-1}{n-1}},\end{aligned}$$

where the second equality uses $\mathbb{P}(S_m=t)=\binom{t-1}{m-1}2^{-t}$, so $\mathbb{P}(S_{n-d}=s-B)/\mathbb{P}(S_n=s)=2^B\binom{s-B-1}{n-d-1}/\binom{s-1}{n-1}$ and the 2^{-B} cancels. A direct computation writes the ratio as

$$\frac{\binom{s-B-1}{n-d-1}}{\binom{s-1}{n-1}}=\prod_{j=1}^{d}(n-j)\cdot\prod_{j=0}^{B-d-1}(s-n-j)\cdot\prod_{j=1}^{B}(s-j)^{-1}.$$

For $s = 2n + O(\sqrt{n})$ each factor in the first product is $n(1 + O(1/n))$, each factor in the second product is $n(1 + O(1/n))$ on the typical event $B = 2d + O(\sqrt{d}\log n)$, and each factor in the third product is $2n(1 + O(1/n))$. Hence

$$\frac{\binom{s-B-1}{n-d-1}}{\binom{s-1}{n-1}} \cdot 2^B = \frac{n^d \cdot n^{B-d}}{(2n)^B} \cdot 2^B \cdot \big(1 + O(d^2/n + |s-2n|d/n)\big) = 1 + O\Big(\frac{d^2 + |s-2n|d}{n}\Big)$$

on the typical event $B = 2d + O(\sqrt{d}\log n)$, recorded as Lemma 3.3 below. Hence the conditional probability tends to $2^{-B} = \prod_i 2^{-b_i}$, which is the i.i.d. law. A standard tail estimate for B (geometric tail of Geom(2)) shows the non-typical contribution is $o(1)$, so the conditional law converges in total variation.

Since $\mathbf{e}_{3^d}(\eta F_d)$ is a bounded continuous function of the finite tuple, the limiting expectation is $\mathbb{E}[\mathbf{e}_{3^d}(\eta F_d(A_1, \dots, A_d))]$.

For $d = 1$ and $\eta = 1$ we have $F_1(A) = 2^{-A}$. Since $2^{-1} \equiv 2 \pmod 3$ and $2^{-2} \equiv 1 \pmod 3$, the powers $2^{-A} \pmod 3$ are 2 if A is odd and 1 if A is even. Under $A \sim \mathrm{Geom}(2)$, $\mathbb{P}(A \text{ odd}) = 2/3$ and $\mathbb{P}(A \text{ even}) = 1/3$. The limit is $\frac{2}{3}\, e^{4\pi i/3} + \frac{1}{3}\, e^{2\pi i/3}$ and a direct computation gives modulus $1/\sqrt{3}$. □

Lemma 3.3 (Composition ratio asymptotics)**.** *For positive integers s, n, d, B with $1 \le d < n$ and $d \le B \le s - n + d - 1$,*

$$\frac{\binom{s-B-1}{n-d-1}}{\binom{s-1}{n-1}} = \frac{\prod_{j=1}^{d}(n-j) \cdot \prod_{j=0}^{B-d-1}(s-n-j)}{\prod_{j=1}^{B}(s-j)}.$$

For $s = 2n + \sigma$ with $|\sigma| = O(\sqrt{n})$, $d = o(\sqrt{n})$, and $B = 2d + O(\sqrt{d\log n})$,

$$\frac{\binom{s-B-1}{n-d-1}}{\binom{s-1}{n-1}} \cdot 2^B = 1 + O\Big(\frac{d^2 + d\log n}{n} + \frac{|\sigma|\sqrt{d\log n}}{n} + \frac{d\sigma^2}{n^2}\Big).$$

In particular, if $|\sigma| = O(\sqrt{n})$ and $d = o(\sqrt{n})$, the right-hand side is $1 + o(1)$.

Proof. The exact form follows from $\binom{a}{b} = a!/(b!(a-b)!)$ after substituting $a = s - B - 1$, $b = n - d - 1$ in the numerator and $a = s - 1$, $b = n - 1$ in the denominator and cancelling factorials.

For the asymptotic, take logarithms. The numerator contributes $\sum_{j=1}^{d}\log(n-j) + \sum_{j=0}^{B-d-1}\log(s-n-j)$ and the denominator contributes $\sum_{j=1}^{B}\log(s-j) + B\log 2$ (after multiplication by 2^B). Writing each factor as $n+O(1)$, $s-n+O(1)$, $s+O(1)$, expanding $\log(N+\delta) = \log N + \delta/N + O(\delta^2/N^2)$, and using $s = 2n + \sigma$:

$$\begin{aligned}\log\frac{\binom{s-B-1}{n-d-1}}{\binom{s-1}{n-1}} + B\log 2 &= d\log n + (B-d)\log(s-n) - B\log s + B\log 2 + O\Big(\frac{d+B}{n}\Big)\\ &= d\log n + (B-d)\log(n+\sigma) - B\log(2n+\sigma) + B\log 2 + O\Big(\frac{d+B}{n}\Big).\end{aligned}$$

Expanding $\log(n+\sigma) = \log n + \sigma/n + O(\sigma^2/n^2)$ and $\log(2n+\sigma) = \log(2n) + \sigma/(2n) + O(\sigma^2/n^2)$,

$$= d\log n + (B-d)\Big(\log n + \frac{\sigma}{n}\Big) - B\Big(\log n + \log 2 + \frac{\sigma}{2n}\Big) + B\log 2 + O\Big(\frac{d^2+\sigma^2}{n^2} + \frac{d+B}{n}\Big).$$

The constant terms cancel: $d + (B-d) - B = 0$ and $-B\log 2 + B\log 2 = 0$. The first-order σ-terms give $(B-d)\sigma/n - B\sigma/(2n) = (B/2 - d)\sigma/n$. Under $B = 2d + O(\sqrt{d\log n})$, $|B/2 - d| = O(\sqrt{d\log n})$, so the first-order σ-term is $O(|\sigma|\sqrt{d\log n}/n)$. The quadratic errors from the Taylor expansions and finite products are $O((d^2 + d\log n)/n + d\sigma^2/n^2)$. Exponentiating gives the claimed asymptotic. □

The fixed-depth obstruction tells us that any positive Fourier-decay statement must require $d = d(\xi) \to \infty$.

4. Subdiffusive microcanonical decay

We next prove that beyond fixed depth, but still in the subdiffusive regime $d = o(\sqrt{n})$, Tao's unconditional decay can be transferred to the microcanonical setting.

Theorem 4.1 (Subdiffusive decay). *Let $s = 2n + O(\sqrt{n})$ and $d = d(n) \to \infty$ with $d = o(\sqrt{n})$. For any $A > 0$,*

$$\sup_{\eta \in (\mathbb{Z}/3^d\mathbb{Z})^\times} \left| \mathbb{E}[\mathbf{e}_{3^d}(\eta F_d(a_{n-d+1}, \dots, a_n)) \mid S_n = s] \right| \le C_A\, d^{-A} + o(1).$$

In particular for $\xi = \eta\, 3^{n+k-d}$ with η as above, $\mathbb{E}[\mathbf{e}_{3^{n+k}}(\xi F_n) \mid S_n = s] \to 0$.

Proof. Tail reduction. By Lemma 3.1, $\mathbf{e}_{3^d}(\eta F_n) = \mathbf{e}_{3^d}(\eta F_d(a_{n-d+1}, \dots, a_n))$, so the conditional expectation depends only on the joint law of the last d variables.

Equivalence of ensembles. Lemma 3.3 (applied with the same d) gives

$$\mathbb{P}\big(a_{n-d+1} = b_1, \dots, a_n = b_d \mid S_n = s\big) = 2^{-B}\big(1 + O(d^2/n + |\sigma| d/n)\big) \tag{15}$$

on the typical event $B = 2d + O(\sqrt{d}\log n)$, where $\sigma := s - 2n$. Under the assumptions $|s - 2n| = O(\sqrt{n})$ and $d = o(\sqrt{n})$,

$$d^2/n + |\sigma| d/n \le d^2/n + \sqrt{n} \cdot d/n = O\big(d^2/n + d/\sqrt{n}\big) = o(1).$$

The non-typical event has $\mathrm{Geom}(2)^{\otimes d}$-mass $O(\exp(-c\log^2 n)) = o(1)$ by Hoeffding's inequality applied to $B - 2d$. Therefore

$$\big\| \mathcal{L}((a_{n-d+1}, \dots, a_n) \mid S_n = s) - \mathrm{Geom}(2)^{\otimes d} \big\|_{\mathrm{TV}} \to 0. \tag{16}$$

Transfer to Tao's bound. Since $\mathbf{e}_{3^d}(\eta F_d)$ has modulus 1, the total-variation estimate (16) gives

$$\mathbb{E}[\mathbf{e}_{3^d}(\eta F_d(a_{n-d+1}, \dots, a_n)) \mid S_n = s] = \mathbb{E}[\mathbf{e}_{3^d}(\eta F_d(A_1, \dots, A_d))] + o(1), \tag{17}$$

where $(A_i)_{i=1}^d$ are i.i.d. Geom(2).

The right-hand side of (17) is precisely the unconditional characteristic function of the Syracuse random variable $\mathrm{Syrac}(\mathbb{Z}/3^d\mathbb{Z}) \equiv F_d(A_1, \dots, A_d) \pmod{3^d}$ at primitive frequency $\eta \in (\mathbb{Z}/3^d\mathbb{Z})^\times$. By Tao's superpolynomial decay (3),

$$\sup_{\eta \in (\mathbb{Z}/3^d\mathbb{Z})^\times} |\mathbb{E}[\mathbf{e}_{3^d}(\eta F_d(A_1, \dots, A_d))]| \le C_A d^{-A}.$$

Combining the two estimates yields the theorem. $\square$

5. Hard-range decay at logarithmic oversampling

We now address the regime $h = k - v_3(\xi) = O(\log n)$. Throughout this section we use Tao's pair-grouping setup [11, §7]. Reverse the variables: set $c_i := a_{n+1-i}$ so that

$$F_n = \sum_{j=1}^{n} 3^{j-1}\, 2^{-c_{[1,j]}}, \qquad c_{[1,j]} := c_1 + \cdots + c_j. \tag{18}$$

For $j \in [n/2]$ set $b_j := c_{2j-1} + c_{2j}$. The b_j are i.i.d. Pascal random variables on $\{2, 3, \dots\}$ with $\mathbb{P}(b_j = b) = (b-1)/2^b$. A short calculation gives

$$F_n = \sum_{j=1}^{\lfloor n/2 \rfloor} 3^{2j-2}\, 2^{-b_{[1,j]}} \big(2^{c_{2j}} + 3\big) \quad (+\,\text{boundary term if } n \text{ is odd}), \tag{19}$$

where $b_{[1,j]} = b_1 + \cdots + b_j$, and conditionally on (b_j) the variables c_{2j} are independent and uniform on $\{1, \dots, b_j - 1\}$. For a character $\chi(x) = e^{-2\pi i \xi x/3^N}$ on $\mathbb{Z}[1/2]$, set

$$f_\chi(x, b) := \mathbb{E}[\chi(x(2^{a_2} + 3)) \mid a_1 + a_2 = b], \qquad |f_\chi(x, b)| \le 1. \tag{20}$$

Tao's Lemma 7.2 [11] gives the cosine cancellation $|f_\chi(x, 3)| = |\cos(\pi\theta(j, b_{[1,j]}))|$, where with $N \ge 1$ and primitive $\eta \in (\mathbb{Z}/3^N\mathbb{Z})^\times$,

$$\theta_{N,\eta}(j, \ell) := \left\{ \frac{\eta \cdot 3^{2j-2} \cdot (2^{-\ell+1} \bmod 3^N)}{3^N} \right\} \in (-1/2, 1/2], \tag{21}$$

and $\{\cdot\}$ denotes the signed fractional part. Tao defines white points of $[N/2] \times \mathbb{Z}$ as those (j, ℓ) with $|\theta_{N,\eta}(j,\ell)| > \varepsilon$ for a fixed small $\varepsilon > 0$, and proves [11, Lemma 7.2] that $b_j = 3$ at a white point implies $|f_\chi(x_j, 3)| \le e^{-\varepsilon^3}$ where $x_j = 3^{2j-2} 2^{-b_{[1,j]}}$.

Define the white-point count

$$W_{n,h,\eta} := \#\big\{1 \le j \le \lfloor n/2 \rfloor : b_j = 3,\ (j, b_{[1,j]}) \text{ is white at modulus } 3^{n+h}\big\}. \tag{22}$$

The pair-grouping computation (19) and (20) give

$$\big|\mathbb{E}[\mathbf{e}_{3^{n+h}}(\eta F_n)]\big| \le \mathbb{E}\, e^{-\varepsilon^3 W_{n,h,\eta}}. \tag{23}$$

Theorem 5.1 (Hard-range decay at logarithmic oversampling)**.** *Fix $K, L, A > 0$. Suppose $|s - 2n| \le L\sqrt{n \log n}$. Then*

$$\sup_{\substack{\xi \in \mathbb{Z}/3^{n+k}\mathbb{Z} \\ 1 \le k - v_3(\xi) \le K \log n}} \big|\mathbb{E}[\mathbf{e}_{3^{n+k}}(\xi F_n) \mid S_n = s]\big| \ll_{A,K,L} n^{-A}.$$

Proof. Write $\xi = 3^v \eta$ with η primitive at modulus 3^{n+h}, $h = k - v$. By (23) applied at modulus $3^{n+h} = 3^N$, $|\mathbb{E}[\mathbf{e}_{3^N}(\eta F_n)]| \le \mathbb{E} e^{-\varepsilon^3 W_{n,h,\eta}}$. We use the standard naive conditioning bound

$$\mathbb{E}[e^{-\varepsilon^3 W} \mid S_n = s] \le \frac{\mathbb{E}[e^{-\varepsilon^3 W}]}{\mathbb{P}(S_n = s)}. \tag{24}$$

The local central limit theorem for S_n (a sum of n i.i.d. Geom(2) variables, $\mathbb{E}S_n = 2n$, $\mathrm{Var} S_n = 2n$) gives

$$\mathbb{P}(S_n = s) = \frac{1}{\sqrt{4\pi n}}\, e^{-(s-2n)^2/(4n)} \big(1 + o(1)\big). \tag{25}$$

For $|s - 2n| \le L\sqrt{n \log n}$ this yields $\mathbb{P}(S_n = s) \gg n^{-1/2 - L^2/4}$. We now produce a polynomial bound for $\mathbb{E} e^{-\varepsilon^3 W_{n,h,\eta}}$.

Augmentation. Tao's Proposition 7.3 [11] controls the white-point count over the full range $[N/2]$ at modulus 3^N, with N Pascal variables. We have only $\lfloor n/2 \rfloor$ Pascal variables. Augment the sequence $b_1, \ldots, b_{\lfloor n/2 \rfloor}$ by sampling $b_{\lfloor n/2 \rfloor + 1}, \ldots, b_{\lfloor N/2 \rfloor}$ as i.i.d. Pascal variables independent of the original sequence. This is purely a coupling device: the augmenting variables do not appear in F_n, but they generate a longer renewal walk for which Tao's estimate applies directly.

Define

$$W^{\text{full}}_{N,\eta} := \#\big\{1 \le j \le \lfloor N/2 \rfloor : b_j = 3,\ (j, b_{[1,j]}) \text{ is white at modulus } 3^N\big\}$$

and $W^{\text{extra}} := W^{\text{full}}_{N,\eta} - W_{n,h,\eta} \ge 0$. Each indicator counted by W^{extra} takes value 0 or 1, so

$$W^{\text{extra}} \le \lfloor N/2 \rfloor - \lfloor n/2 \rfloor \le h \tag{26}$$

deterministically. From $W^{\text{full}}_{N,\eta} = W_{n,h,\eta} + W^{\text{extra}}$ and (26),

$$e^{-\varepsilon^3 W_{n,h,\eta}} = e^{-\varepsilon^3 W^{\text{full}}_{N,\eta}} \cdot e^{\varepsilon^3 W^{\text{extra}}} \le e^{\varepsilon^3 h} \cdot e^{-\varepsilon^3 W^{\text{full}}_{N,\eta}} \quad \text{pointwise.} \tag{27}$$

Tao's Proposition 7.3 [11, §7.5] gives, for any $A > 0$,

$$\mathbb{E}\, e^{-\varepsilon^3 W^{\text{full}}_{N,\eta}} \ll_A N^{-A}, \tag{28}$$

uniformly in $N \ge 1$ and primitive $\eta \in (\mathbb{Z}/3^N\mathbb{Z})^\times$. Taking expectations of (27) (the augmenting variables have the same Pascal law as the originals, so the joint expectation is unchanged),

$$\mathbb{E}\, e^{-\varepsilon^3 W_{n,h,\eta}} \le e^{\varepsilon^3 h}\, \mathbb{E}\, e^{-\varepsilon^3 W^{\text{full}}_{N,\eta}} \ll_A e^{\varepsilon^3 h}\, N^{-A}. \tag{29}$$

Polynomial absorption. For $h \le K \log n$ we have $e^{\varepsilon^3 h} \le n^{\varepsilon^3 K}$ and $N = n + h \le n(1 + K \log n / n)$, so $\log N = \log n + o(1)$ and $N^{-A} = n^{-A}(1 + o(1))$. Substituting into (29),

$$\mathbb{E}\, e^{-\varepsilon^3 W_{n,h,\eta}} \ll_{A,K} n^{\varepsilon^3 K} \cdot n^{-A} = n^{-(A - \varepsilon^3 K)}.$$

Combining with (24) and the local CLT bound $\mathbb{P}(S_n = s)^{-1} \ll_L n^{1/2+L^2/4}$ valid for $|s - 2n| \le L\sqrt{n \log n}$,

$$\left|\mathbb{E}[\mathbf{e}_{3^N}(\eta F_n) \mid S_n = s]\right| \le \mathbb{E}[e^{-\varepsilon^3 W_{n,h,\eta}} \mid S_n = s] \ll_{A,K,L} n^{-(A-\varepsilon^3 K - 1/2 - L^2/4)}.$$

Choosing A sufficiently large (depending on the desired exponent A', K, L) yields the bound $\ll_{A',K,L} n^{-A'}$.

The factorization (4) reduces the affine character at $\xi = 3^v \eta$ to the offset character at modulus 3^N: $\mathbf{e}_{3^{n+k}}(\xi \mathrm{Aff}_{a_1,\dots,a_n}(x)) = \mathbf{e}_{3^k}(\xi\, 2^{-s} x) \cdot \mathbf{e}_{3^N}(\eta F_n)$. Under $S_n = s$ the multiplicative factor is deterministic of modulus 1, so the conditional expectation over $(a_1, \dots, a_n)$ inherits the same bound. □

6. Bernoulli-bridge normal form

The proof of Theorems 1.2 and 1.4 relies on rewriting the offset F_n in a Bernoulli-bridge form that collapses the 3-adic phase to a real-line integral against a perpetuity.

Set $m := s - 1$, $r := n - 1$. Under $S_n = s$ all compositions $a_1 + \cdots + a_n = s$, $a_i \ge 1$, are equiprobable. The standard stars-and-bars bijection identifies such compositions with binary strings $\varepsilon = (\varepsilon_1, \dots, \varepsilon_m)$ having exactly r ones; the position of the t-th one is $p_t = b_1 + \cdots + b_t$ where we write the reverse composition $b_i = a_{n+1-i}$. Let $\Omega_{m,r} := \{\varepsilon \in \{0,1\}^m : \sum_j \varepsilon_j = r\}$, $R_j := \varepsilon_1 + \cdots + \varepsilon_j$,

$$X_m := \sum_{j=1}^{m} \varepsilon_j\, 3^{R_{j-1}}\, 2^{-j}, \qquad P_m := 2^m X_m = \sum_{j=1}^{m} \varepsilon_j\, 3^{R_{j-1}}\, 2^{m-j} \in \mathbb{Z}. \tag{30}$$

Lemma 6.1 (Decomposition of F_n)**.** *Under $S_n = s$, $F_n = 3^{n-1}\, 2^{-s} + X_m$.*

Proof. Reversal. Substitute $m' = n - m + 1$ in (2), so $m = n - m' + 1$ and $n - m = m' - 1$; the range $m = 1, \dots, n$ becomes $m' = n, \dots, 1$ (i.e. all of $\{1, \dots, n\}$). With $b_i = a_{n+1-i}$, $a_{[m,n]} = a_m + \cdots + a_n = a_{n-m'+1} + \cdots + a_n = b_{m'} + \cdots + b_1 = b_{[1,m']}$. Therefore

$$F_n = \sum_{m'=1}^{n} 3^{m'-1}\, 2^{-b_{[1,m']}} = \sum_{\ell=0}^{n-1} 3^{\ell}\, 2^{-b_{[1,\ell+1]}}. \tag{31}$$

The term $\ell = n - 1$ contributes $3^{n-1}\, 2^{-b_{[1,n]}} = 3^{n-1}\, 2^{-s}$.

Stars-and-bars bijection. Identify positive compositions $(b_1, \dots, b_n)$ of s with binary strings $\varepsilon \in \{0,1\}^{s-1}$ having exactly $n-1$ ones as follows. Read ε from left to right; let $p_1 < p_2 < \cdots < p_{n-1}$ be the positions of the ones. Set $p_0 = 0$, $p_n = s$, and define $b_t := p_t - p_{t-1}$ for $t = 1, \dots, n$. Each $b_t \ge 1$ and $\sum b_t = s$, so $\varepsilon \mapsto (b_1, \dots, b_n)$ is a bijection between $\Omega_{m,r} = \{\varepsilon \in \{0,1\}^m : \sum = r\}$ and positive compositions, where $m = s - 1$, $r = n - 1$. Moreover the partial sum $b_{[1,t]} = p_t$ is exactly the position of the t-th one. At position $j \in \{1, \dots, m\}$ in the Bernoulli string, $\varepsilon_j = 1$ iff j is the position of some p_t, in which case $t = R_j$ (the running one-count at j).

Identifying X_m. For $0 \le \ell \le n-2$, the term $3^\ell\, 2^{-b_{[1,\ell+1]}}$ in (31) corresponds to $j = p_{\ell+1}$ in the Bernoulli string, with $R_{j-1} = \ell$ (since $p_{\ell+1}$ is the position of the $(\ell+1)$-st one). Summing over $\ell = 0, \dots, n-2$ and recognising $\varepsilon_j \cdot 3^{R_{j-1}} \cdot 2^{-j}$ as the contribution at each position with $\varepsilon_j = 1$,

$$\sum_{\ell=0}^{n-2} 3^\ell\, 2^{-b_{[1,\ell+1]}} = \sum_{j:\,\varepsilon_j=1} 3^{R_{j-1}}\, 2^{-j} = \sum_{j=1}^{m} \varepsilon_j\, 3^{R_{j-1}}\, 2^{-j} = X_m.$$

Combining with the $\ell = n - 1$ contribution gives $F_n = 3^{n-1}\, 2^{-s} + X_m$. □

Recall that $h \ge 1$, $N = n + h$, and $\eta \in (\mathbb{Z}/3^N\mathbb{Z})^\times$. Let $q \in \mathbb{Z}$ be the unique representative of $\eta \cdot 2^{-s} \bmod 3^N$ in $(-3^N/2, 3^N/2]$. Since η is a unit and 2 is invertible modulo 3^N, q is also coprime to 3:

$$3 \nmid q. \tag{32}$$

Theorem 6.2 (Bridge reduction). *For every $\eta \in (\mathbb{Z}/3^N\mathbb{Z})^\times$,*

$$\mathbb{E}[\mathbf{e}_{3^N}(\eta F_n) \mid S_n = s] = \mathbf{e}_{3^N}(q \cdot 3^{n-1}) \cdot \mathbb{E}_{\Omega_{m,r}} \mathbf{e}_{3^N}(2qP_m). \tag{33}$$

For the resonant frequency $\eta = 2^s$, $q = 1$, the formula simplifies to $\mathbf{e}_{3^N}(3^{n-1}) \cdot \mathbb{E}_{\Omega_{m,r}} \mathbf{e}_{3^N}(2P_m)$.

Proof. By Lemma 6.1, $F_n = 3^{n-1}\,2^{-s} + X_m$ in $\mathbb{Z}[1/2] \cap (\mathbb{Z}/3^N\mathbb{Z})$ (using the canonical embedding induced by 2 being a unit modulo 3^N). Multiplying by $\eta \in (\mathbb{Z}/3^N\mathbb{Z})^\times$,

$$\eta F_n \equiv \eta \cdot 3^{n-1} \cdot 2^{-s} + \eta X_m \pmod{3^N}.$$

By definition $q \equiv \eta \cdot 2^{-s} \pmod{3^N}$, hence $\eta \equiv q \cdot 2^s \pmod{3^N}$ and $\eta \cdot 2^{-s} \equiv q \pmod{3^N}$. Substituting,

$$\eta F_n \equiv q \cdot 3^{n-1} + q \cdot 2^s \cdot X_m \pmod{3^N}. \tag{34}$$

Since $m = s-1$, $2^s = 2 \cdot 2^m$, so $2^s X_m = 2 \cdot 2^m X_m = 2P_m \in \mathbb{Z}$. Therefore $\eta F_n \equiv q \cdot 3^{n-1} + 2qP_m \pmod{3^N}$, both terms being integers.

Under $S_n = s$, the law of $(a_1, \dots, a_n)$ is uniform on the set of positive compositions of s into n parts, which has size $\binom{s-1}{n-1} = \#\Omega_{m,r}$. The stars-and-bars bijection in the proof of Lemma 6.1 identifies compositions with $\Omega_{m,r}$, so the law of ε, hence the law of $P_m = 2^m X_m$, is uniform on $\Omega_{m,r}$. The constant phase $\mathbf{e}_{3^N}(q \cdot 3^{n-1})$ pulls out of the conditional expectation, leaving $\mathbb{E}[\mathbf{e}_{3^N}(\eta F_n) \mid S_n = s] = \mathbf{e}_{3^N}(q \cdot 3^{n-1}) \cdot \mathbb{E}_{\Omega_{m,r}} \mathbf{e}_{3^N}(2qP_m)$.

For the resonant frequency $\eta = 2^s$, $q \equiv 2^s \cdot 2^{-s} \equiv 1 \pmod{3^N}$, so $q = 1$ in $(-3^N/2, 3^N/2]$.

Primitivity. Since $\eta \in (\mathbb{Z}/3^N\mathbb{Z})^\times$ and 2^{-s} is a unit modulo 3^N, their product q is also a unit, in particular $3 \nmid q$. □

Lemma 6.3 (Multiscale anti-resonance). *Let $t = q\,2^s/3^N$ with $3 \nmid q$, $N = n+h$, $h \geq 1$, $m = s-1$. Set $\Delta := s \log_3 2 - N = s \log_3 2 - n - h$ and*

$$\mathfrak{D}_m(t) := \min_{0 \leq \ell \leq m} 2^\ell \cdot \operatorname{dist}(t, 2^{m-\ell}\mathbb{Z}).$$

Then $\mathfrak{D}_m(t) \geq \frac{1}{2}\,3^\Delta$.

Proof. For any integer A and $\ell \in \{0, \dots, m\}$,

$$\begin{aligned} 2^\ell |t - A\,2^{m-\ell}| &= \frac{2^\ell \cdot 2^{m-\ell}}{3^N} |q\,2^{s-(m-\ell)} - A\,3^N| \\ &= \frac{2^{s-1}}{3^N} |2^{\ell+1} q - A\,3^N|. \end{aligned}$$

Since $\gcd(q,3) = 1 = \gcd(2,3)$, we have $2^{\ell+1} q \not\equiv 0 \pmod 3$ while $A\,3^N \equiv 0 \pmod 3$, so $2^{\ell+1}q - A\,3^N$ is a non-zero integer of absolute value ≥ 1. Therefore $2^\ell \operatorname{dist}(t, 2^{m-\ell}\mathbb{Z}) \geq 2^{s-1}/3^N = \frac{1}{2}\,3^\Delta$. □

The Bernoulli-bridge normal form of Theorem 6.2 suggests considering the unconditional perpetuity obtained by passing to the iid fair-Bernoulli limit. Let $\varepsilon_1, \varepsilon_2, \dots$ be i.i.d. uniform on $\{0,1\}$ and set

$$\mathcal{F}_\infty := \sum_{j=1}^\infty \varepsilon_j\, 3^{R_{j-1}}\, 2^{-j}, \quad R_j = \varepsilon_1 + \cdots + \varepsilon_j. \tag{35}$$

The series converges almost surely because the gaps between successive ones are i.i.d. Geom(2) and the multiplicative drift $\mathbb{E}[\log(3 \cdot 2^{-A})] = \log 3 - 2\log 2 = \log(3/4) < 0$; this is a special case of the random difference equation framework of Vervaat [12].

Lemma 6.4 (Distributional equation). *$\mathcal{F}_\infty \overset{d}{=} 2^{-A}(1 + 3\,\mathcal{F}'_\infty)$, where $A \sim \mathrm{Geom}(2)$ and $\mathcal{F}'_\infty$ is an independent copy of $\mathcal{F}_\infty$. Equivalently, $\mathcal{F}_\infty$ is the unique stationary measure of the two-map IFS*

$$\psi_0(x) = x/2, \qquad \psi_1(x) = 1/2 + (3/2)x, \tag{36}$$

with ψ_0 chosen with probability $1/2$ and ψ_1 with probability $1/2$.

Proof. Decompose $\mathcal{F}_\infty$ by ε_1. If $\varepsilon_1 = 0$, the sum starts at $j = 2$ and a re-indexing gives $\mathcal{F}_\infty = \frac{1}{2}\mathcal{F}'_\infty$. If $\varepsilon_1 = 1$, the $j = 1$ term is $1 \cdot 3^0 \cdot 2^{-1} = 1/2$ and the remaining terms have $R_{j-1} = 1 + (\text{rest})$, giving $\mathcal{F}_\infty = \frac{1}{2} + \frac{3}{2}\mathcal{F}'_\infty$. Each case has probability 1/2. For the geometric reformulation, the gap to the first one is $A := \min\{j : \varepsilon_j = 1\} \sim \mathrm{Geom}(2)$, and $\mathcal{F}_\infty = 2^{-A}(1 + 3\,\mathcal{F}'_\infty)$ follows by aggregating the geometric factor. □

7. Rajchman property of the perpetuity

We now establish that $\widehat{\mu}_{\mathcal{F}}(t) := \mathbb{E}\, e^{2\pi i t \mathcal{F}_\infty}$ vanishes at infinity. This is the key analytic input for the resonant-decay theorems below.

Theorem 7.1 (Brémont's Rajchman criterion, restated)**.** *Let $N \geq 1$, and let $\varphi_k(x) = r_k x + b_k$ $(0 \leq k \leq N)$ be affine maps of the real line with $r_k > 0$. Let $p = (p_0, \ldots, p_N)$ be a probability vector with $p_k > 0$, and assume $\sum_{k=0}^N p_k \log r_k < 0$ (contracting on average) and that $(\varphi_k)_{0\leq k\leq N}$ have no common fixed point. Let ν be the unique stationary measure satisfying $\nu = \sum_k p_k\, (\varphi_k)_*\nu$. Then ν fails to be Rajchman, i.e. $\widehat{\nu}(t) \not\to 0$ as $|t| \to \infty$, if and only if there exists an affine map $f(x) = ax + b$ $(a \neq 0)$ such that the conjugated family $f \circ \varphi_k \circ f^{-1}$ is in* Pisot form*: there exist $0 < \lambda < 1$ with $1/\lambda$ a Pisot number, relatively prime integers $(n_k)_{0\leq k\leq N}$, and translations $\mu_k \in T(1/\lambda)$ such that $f \circ \varphi_k \circ f^{-1}(x) = \lambda^{n_k} x + \mu_k$ for every k. In particular, in the non-Rajchman case the original ratios satisfy $r_k = \lambda^{n_k}$ for one common λ.*

This is [1, Theorem 2.5], with the Pisot form definition from [1, Definition 2.3]. The result generalises the classical Erdős–Salem characterisation of Rajchman Bernoulli convolutions [2, 9]; see [7] for a survey of the Bernoulli-convolution background.

Theorem 7.2 (Rajchman of $\mathcal{F}_\infty$)**.** $\widehat{\mu}_{\mathcal{F}}(t) \to 0$ *as* $|t| \to \infty$.

Proof. We apply Theorem 7.1 (Brémont [1, Theorem 2.5]) and then check that no affine conjugation of the IFS (36) lands in Pisot form.

Verifying hypotheses. The IFS has affine maps $\psi_0(x) = r_0 x + b_0$ and $\psi_1(x) = r_1 x + b_1$ with

$$r_0 = 1/2, \quad b_0 = 0, \quad r_1 = 3/2, \quad b_1 = 1/2, \quad p_0 = p_1 = 1/2.$$

(i) No common fixed point. The fixed point of ψ_0 is $x = 0$. The fixed point of ψ_1 solves $1/2 + (3/2)x = x$, i.e. $-x/2 = 1/2$, giving $x = -1$. These are distinct.

(ii) Contracting on average. The average Lyapunov exponent is

$$p_0 \log r_0 + p_1 \log r_1 = \tfrac{1}{2}\log(1/2) + \tfrac{1}{2}\log(3/2) = \tfrac{1}{2}\log(3/4) < 0.$$

By Theorem 7.1, the invariant measure $\mathcal{F}_\infty$ fails to be Rajchman if and only if some affine conjugate of the IFS (36) is in Pisot form.

Affine conjugation preserves linear ratios. For $f(x) = ax + b$ $(a \neq 0)$ and any $g(x) = rx + c$, $f \circ g \circ f^{-1}(x) = a(r((x-b)/a) + c) + b = rx + (ac - rb + b)$, which has the same linear ratio r as g. Hence the conjugate of the IFS (36) has linear ratios $r_0 = 1/2$ and $r_1 = 3/2$ regardless of f.

Excluding Pisot form. For Pisot form we would need $1/2 = \lambda^{n_0}$ and $3/2 = \lambda^{n_1}$ for some Pisot $1/\lambda > 1$ and coprime integers $(n_0, n_1) \in \mathbb{Z}^2$. Brémont's framework allows the integer exponents (n_j) to be of either sign, and $\lambda \in (0,1)$ since $1/\lambda > 1$. Eliminating λ via $\lambda^{n_0 n_1} = (1/2)^{n_1} = (3/2)^{n_0}$,

$$2^{-n_1} = 2^{-n_0} \cdot 3^{n_0} \quad \Longrightarrow \quad 2^{n_0 - n_1} \cdot 3^{-n_0} = 1 \text{ in } \mathbb{Q}^\times.$$

The multiplicative group $\mathbb{Q}^\times$ has the 2-adic and 3-adic valuations v_2, v_3 as independent characters. Applying v_2 gives $n_0 - n_1 = 0$; applying v_3 gives $-n_0 = 0$, hence $n_0 = 0$ and $n_1 = 0$. This contradicts $\gcd(n_0, n_1) = 1$ (which requires $(n_0, n_1) \neq (0,0)$).

Therefore the IFS (36) is not affinely conjugate to any Pisot form, and Brémont's theorem yields the Rajchman property of $\mathcal{F}_\infty$. □

Remark 7.3 (Why the perpetuity matters)**.** $\mathcal{F}_\infty$ is the natural real-line limit object that arises from the resonant frequency $\eta_s = 2^s$ via the bridge reduction (33). The Rajchman property converts a multiplicative obstruction in the 3-adic affine model into a Diophantine question on a real-line perpetuity, where the classical Salem–Erdős/Brémont machinery applies. This is the link that gives a sharp resonant phase transition.

8. Sharp resonant phase transition

8.1. **Near-critical decay.** We first treat the near-critical range, in which the bridge length scale R used for ensemble equivalence can be chosen $o(n/\log n)$. Two auxiliary lemmas prepare the way.

Lemma 8.1 (Long-block ensemble equivalence)**.** *Fix $L_0 > 0$. Let $a_1, a_2, \dots$ be i.i.d.* Geom(2), *$S_n = a_1 + \cdots + a_n$, and assume $|s - 2n| \le L_0\sqrt{n \log n}$. Let $R = R_n$ satisfy $1 \le R \le n/2$ and $R \log n/n \le 1$. Then for every $A > 0$,*

(37)

$$\|\mathcal{L}((a_n, a_{n-1}, \dots, a_{n-R+1}) \mid S_n = s) - \mathrm{Geom}(2)^{\otimes R}\|_{\mathrm{TV}} \le C_{A,L_0}\left[\sqrt{\frac{R\log n}{n}} + \frac{R\log n + \log^2 n}{n} + n^{-A}\right].$$

In particular, if $R \log n/n \to 0$, the total-variation distance tends to 0.

Proof. Write Geom(p) for the law on $\{1, 2, \dots\}$ with $\mathbb{P}_p(X = m) = p(1-p)^{m-1}$, so Geom(2) = Geom(1/2). Set $p := n/s$. For $|s - 2n| \le L_0\sqrt{n \log n}$ and n large, $p \in [1/3, 2/3]$ and $|p - 1/2| \le C_{L_0}\sqrt{\log n/n}$.

Step 1: Conditional law equals tilted-product conditional. Under any product law Geom$(p)^{\otimes n}$, every $(a_1, \dots, a_n)$ with $\sum a_i = s$ has the same mass $p^n(1-p)^{s-n}$, so the conditional law on $\{S_n = s\}$ is independent of p. Hence $\mathcal{L}((a_n, \dots, a_{n-R+1}) \mid S_n = s)$ is the same under Geom(p) as under Geom(1/2). We compare it first with Geom$(p)^{\otimes R}$, then Geom$(p)^{\otimes R}$ with Geom$(1/2)^{\otimes R}$.

Step 2: Tilted local CLT. Let $T_R = a_n + \cdots + a_{n-R+1}$. By independence under Geom(p), the Radon–Nikodym derivative of the conditional block law w.r.t. Geom$(p)^{\otimes R}$ is $L_p(T_R) = \mathbb{P}_p(S_{n-R} = s - T_R)/\mathbb{P}_p(S_n = s)$, so

$$\|\mathcal{L}_{\mathrm{cond}} - \mathrm{Geom}(p)^{\otimes R}\|_{\mathrm{TV}} = \tfrac{1}{2}\mathbb{E}_p|L_p(T_R) - 1|. \tag{38}$$

A negative-binomial Stirling expansion gives, uniformly for $|x| \le m^{2/3}$, $\mathbb{P}_p(S_m^{(p)} = m/p + x) = (2\pi v_p m)^{-1/2}\exp(-x^2/(2v_p m))(1+O((1+|x|+|x|^3)/m))$, where $v_p = (1-p)/p^2$. Set $u := T_R - R/p$; Chernoff for Geom(p) with bounded MGF gives $\mathbb{P}_p(|u| > t) \le 2\exp(-c_0\min(t^2/R, t))$. Choose $t = K_A(\sqrt{R\log n} + \log n)$ for K_A large; then $\mathbb{P}_p(|u| > t) \le n^{-A-2}$. On $\{|u| \le t\}$, the Stirling expansion gives $L_p(T_R) = \sqrt{n/(n-R)}\exp(-u^2/(2v_p(n-R)))(1+O(R/n+t^2/n+t^3/n^2+1/n))$, so $\sup_{|u|\le t}|L_p(T_R) - 1| \le C_{A,L_0}(R\log n + \log^2 n)/n$. Together with $L_p(T_R) \le C$ pointwise (also from Stirling) and $\mathbb{P}_p(|u| > t) \le n^{-A-2}$,

$$\|\mathcal{L}_{\mathrm{cond}} - \mathrm{Geom}(p)^{\otimes R}\|_{\mathrm{TV}} \le C_{A,L_0}\left[\frac{R\log n + \log^2 n}{n} + n^{-A}\right]. \tag{39}$$

Step 3: Tilt-to-untilt via Hellinger affinity. The Hellinger affinity of one Geom(p) vs. one Geom(1/2) is $\rho(p) = \sqrt{p/2}/(1 - \sqrt{(1-p)/2})$. Taylor at $p = 1/2$ gives $1 - \rho(p) \le C(p - 1/2)^2$, so $H^2(\mathrm{Geom}(p)^{\otimes R}, \mathrm{Geom}(1/2)^{\otimes R}) = 1 - \rho(p)^R \le R(1 - \rho(p)) \le CR(p - 1/2)^2$. Total variation $\le \sqrt{2}\cdot H$ gives $\|\mathrm{Geom}(p)^{\otimes R} - \mathrm{Geom}(1/2)^{\otimes R}\|_{\mathrm{TV}} \le C\sqrt{R}\,|p - 1/2| \le C_{L_0}\sqrt{R\log n/n}$.

Combining steps 2 and 3 by the triangle inequality yields (37). □

Lemma 8.2 (Uniform bridge tail moment)**.** *There exist $\alpha \in (0, 1)$, $\rho \in (0, 1)$, and $C < \infty$ such that for $r = m/2 + O(\sqrt{m\log m})$ and $0 \le R \le m$,*

$$\mathbb{E}_{\Omega_{m,r}}\left[\left(\textstyle\sum_{j>R}\varepsilon_j\, 3^{R_{j-1}}\, 2^{-j}\right)^{\alpha}\right] \le C\rho^R.$$

In particular, $\sup_{m,r}\mathbb{E}_{\Omega_{m,r}}[X_m^{\alpha}] < \infty$.

Proof. Choose $p_+ \in (1/2, \log 2/\log 3)$. For m large, $r/m \le p_+$. Pick $\alpha > 0$ small enough that $\rho := (1 - p_+ + p_+ 3^{\alpha})/2^{\alpha} < 1$; this is possible because $\frac{d}{d\alpha}\log\rho|_{\alpha=0} = p_+\log 3 - \log 2 < 0$.

By subadditivity of $x \mapsto x^{\alpha}$ on $\mathbb{R}_{\ge 0}$,

$$\Big(\sum_{j>R}\varepsilon_j\, 3^{R_{j-1}}2^{-j}\Big)^{\alpha} \le \sum_{j>R}\varepsilon_j\, 3^{\alpha R_{j-1}}2^{-\alpha j}.$$

Conditional on $\varepsilon_j = 1$, R_{j-1} is hypergeometric (drawing $j-1$ from $m-1$ with $r-1$ marked). Hoeffding's convex-order comparison for sampling without replacement gives, for any $a \geq 1$, $\mathbb{E}[a^{R_{j-1}} \mid \varepsilon_j = 1] \leq (1 - p_j + p_j a)^{j-1}$ with $p_j = (r-1)/(m-1) \leq p_+ + o(1)$. Taking $a = 3^\alpha$ and absorbing the $o(1)$,

$$\mathbb{E}_{\Omega_{m,r}}\left[\varepsilon_j\, 3^{\alpha R_{j-1}} 2^{-\alpha j}\right] \leq \frac{r}{m} \cdot 2^{-\alpha j}(1 - p_+ + p_+\, 3^\alpha)^{j-1} \leq C\rho^j.$$

Summing in $j > R$ gives the claim. The bound on $\mathbb{E}[X_m^\alpha]$ is the case $R = 0$. $\square$

Theorem 8.3 (Near-critical resonant decay)**.** *Let $s = 2n + O(\sqrt{n \log n})$, $h \geq 1$, and define $\Delta_n := s \log_3 2 - n - h$. Suppose $\Delta_n \to +\infty$ and $\Delta_n = o(n/\log n)$. For the resonant frequency $\eta_s = 2^s \bmod 3^{n+h}$,*

$$\mathbb{E}[\mathbf{e}_{3^{n+h}}(2^s F_n) \mid S_n = s] \to 0.$$

Proof. By Theorem 6.2 with $q = 1$, the conditional expectation equals (up to a deterministic phase) $\mathbb{E}_{\Omega_{m,r}} \exp(2\pi i \lambda_n X_m)$ with $\lambda_n = 3^{\Delta_n} \to \infty$ and $m = s-1$, $r = n-1$.

Choose $R = R_n$ with $\Delta_n + \log n \ll R \ll n/\log n$; this is feasible because $\Delta_n = o(n/\log n)$. Set $X_R := \sum_{j=1}^R \varepsilon_j 3^{R_{j-1}} 2^{-j}$ and $T_R := X_m - X_R$.

By Lemma 8.2, $\mathbb{E}_{\Omega_{m,r}}[T_R^\alpha] \leq C\rho^R$ for the absolute α, ρ given there. Using $|e^{ix} - 1| \leq C_\alpha |x|^\alpha$ on $\mathbb{R}_{\geq 0}$ for $0 < \alpha \leq 1$,

$$\left|\mathbb{E}_{\Omega_{m,r}} e^{2\pi i \lambda_n X_m} - \mathbb{E}_{\Omega_{m,r}} e^{2\pi i \lambda_n X_R}\right| \leq C_\alpha\, \lambda_n^\alpha\, \mathbb{E}_{\Omega_{m,r}}[T_R^\alpha] \leq C_\alpha\, 3^{\alpha \Delta_n}\, \rho^R = o(1),$$

by the choice $R \gg \Delta_n + \log n$.

By Lemma 8.1 (applied with $R \log n/n \to 0$ since $R \ll n/\log n$), the joint law of $(A_0, \ldots, A_{R-1}) = (a_n, \ldots, a_{n-R+1})$ under $S_n = s$ is close in total variation to $\mathrm{Geom}(2)^{\otimes R}$. Since X_R depends only on $(A_0, \ldots, A_{R-1})$,

$$\mathbb{E}_{\Omega_{m,r}} e^{2\pi i \lambda_n X_R} = \mathbb{E}_{\mathrm{iid}} e^{2\pi i \lambda_n X_R} + o(1).$$

Applying the same tail bound under the i.i.d. law to the perpetuity $\mathcal{F}_\infty = \sum_{j \geq 1} \varepsilon_j 3^{R_{j-1}} 2^{-j}$, $\mathbb{E}_{\mathrm{iid}} |\mathcal{F}_\infty - X_R|^\alpha \ll \rho^R$, hence $|\mathbb{E}_{\mathrm{iid}} e^{2\pi i \lambda_n X_R} - \widehat{\mu}_{\mathcal{F}}(\lambda_n)| \leq C_\alpha\, 3^{\alpha \Delta_n}\, \rho^R = o(1)$. By Theorem 7.2, $\widehat{\mu}_{\mathcal{F}}(\lambda_n) \to 0$ since $\lambda_n \to \infty$. Combining gives $\mathbb{E}_{\Omega_{m,r}} \exp(2\pi i \lambda_n X_m) \to 0$, and hence the stated conditional expectation tends to 0. $\square$

8.2. **Suffix self-similarity and full subcritical decay.** The near-critical theorem covers $\Delta_n = o(n/\log n)$. For larger Δ_n we use an exact suffix self-similarity to reduce to a smaller bridge whose entropy gap is $o(L/\log L)$.

Lemma 8.4 (Suffix scale selection)**.** *Let $s = 2n + O(\sqrt{n \log n})$ and $\Delta_n := s \log_3 2 - n - h$. Assume $\Delta_n \geq c_0\, n/\log n$ for some fixed $c_0 > 0$, and $h \to \infty$. Put $\beta := \log_3 2 - 1/2 > 0$. Then $R_n := \lfloor h^{3/4}/(\log h)^{3/4} \rfloor$ satisfies $\sqrt{h \log h} \ll R_n \ll h/\log^2 h$ and $R_n \ll \Delta_n$. With $L := \lfloor (h + R_n)/\beta \rfloor$, one has $L \asymp h$, $R_n = o(L/\log^2 L)$, and $R_n \gg \sqrt{L \log L}$.*

Proof. The inequalities $\sqrt{h \log h} \ll h^{3/4}/(\log h)^{3/4} \ll h/\log^2 h$ are both equivalent to $(\log h)^5 \ll h$, which holds for $h \to \infty$. Since $\Delta_n \geq c_0 n/\log n$ and $h \leq s \log_3 2 - n + O(1) = O(n)$, $R_n \leq n^{3/4} \ll c_0 n/\log n \leq \Delta_n$. The claims about L follow from $L = (h + R_n)/\beta + O(1)$ and $R_n = o(h)$. $\square$

Lemma 8.5 (Suffix decomposition)**.** *Let $1 \leq J < m$, $L = m - J$, and let $\omega := (\varepsilon_{J+1}, \ldots, \varepsilon_m)$ with suffix sum $y := \omega_1 + \cdots + \omega_L$. Then*

$$X_m = X_J + 3^{r-y} \cdot 2^{-J} \cdot X_L(\omega),$$

where $X_L(\omega) = \sum_{i=1}^L \omega_i\, 3^{\omega_1 + \cdots + \omega_{i-1}}\, 2^{-i}$ is the bridge sum on the suffix.

Proof. Split X_m at index J. For $J < j \leq m$, write $i = j - J$ and observe $R_{j-1} = R_J + \omega_1 + \cdots + \omega_{i-1} = (r-y) + \omega_1 + \cdots + \omega_{i-1}$. Therefore the suffix sum equals $3^{r-y} \cdot 2^{-J} \sum_{i=1}^L \omega_i 3^{\omega_1 + \cdots + \omega_{i-1}} 2^{-i}$. $\square$

Theorem 8.6 (Sharp resonant phase transition)**.** *Theorem 1.2 holds. That is, for $s = 2n + O(\sqrt{n \log n})$ and $h \geq 1$:*

(1) *if* $\Delta_n \to +\infty$*, the conditional Fourier coefficient* $\mathbb{E}[\mathbf{e}_{3^{n+h}}(2^s F_n) \mid S_n = s] \to 0$*;*
(2) *if* $\Delta_n \to -\infty$*,* $|\mathbb{E}[\,\cdot\,]| \to 1$*;*
(3) *if* $3^{\Delta_n} \to \lambda \in (0,\infty)$*, the coefficient tends to* $\mathbf{e}_{3^{n+h}}(3^{n-1})\,\widehat{\mu}_{\mathcal{F}}(\lambda)$*.*

Lemma 8.7 (Uniform L^α-tightness under the central bridge)**.** *With* α, ρ, C *as in Lemma 8.2, uniformly for* $s = 2n + O(\sqrt{n\log n})$*,* $\sup_n \mathbb{E}[(F_n^{\mathbb{R}})^\alpha \mid S_n = s] \le C'$ *for some* $C' < \infty$*. In particular* $F_n^{\mathbb{R}} \mid S_n = s$ *is tight.*

Proof. By Lemma 6.1, $F_n^{\mathbb{R}} = 3^{n-1}2^{-s} + X_m$ under the bridge representation $\Omega_{m,r}$, with $m = s-1$, $r = n-1$. Since $s = 2n + O(\sqrt{n\log n})$, $r/m = 1/2 + O(\sqrt{\log n/n})$, so the hypothesis of Lemma 8.2 is met. That lemma gives $\mathbb{E}_{\Omega_{m,r}}[X_m^\alpha] \le C$. The deterministic term satisfies $3^{n-1}2^{-s} \ll (3/4)^n e^{O(\sqrt{n\log n})} = o(1)$, hence $\mathbb{E}[(F_n^{\mathbb{R}})^\alpha \mid S_n = s] \le C_\alpha((3^{n-1}2^{-s})^\alpha + \mathbb{E}_{\Omega_{m,r}}[X_m^\alpha]) \le C'$. Markov's inequality $\mathbb{P}(F_n^{\mathbb{R}} > M \mid S_n = s) \le C' M^{-\alpha}$ yields tightness. □

Proof of Theorem 8.6. We treat the three parts in turn.

Part (3): critical profile. With $\lambda_n = 3^{\Delta_n} \to \lambda \in (0,\infty)$, the truncation argument of Theorem 8.3 (with $R \gg \log n$, here $\Delta_n = O(1)$) gives $\mathbb{E}[\exp(2\pi i \lambda_n F_n^{\mathbb{R}}) \mid S_n = s] = \widehat{\mu}_{\mathcal{F}}(\lambda_n) + o(1) \to \widehat{\mu}_{\mathcal{F}}(\lambda)$ by continuity of $\widehat{\mu}_{\mathcal{F}}$ on the compact set $\{|t| \le |\lambda| + 1\}$. Multiplying by the deterministic phase $\mathbf{e}_{3^{n+h}}(3^{n-1})$ gives part (3).

Part (2): supercritical obstruction. With $\lambda_n = 3^{\Delta_n} \to 0$, Lemma 8.7 gives a uniform L^α bound for $F_n^{\mathbb{R}} \mid S_n = s$. For any $\delta > 0$, Markov's inequality gives $\mathbb{P}(\lambda_n F_n^{\mathbb{R}} > \delta \mid S_n = s) \le \delta^{-\alpha}\lambda_n^\alpha C' \to 0$. Hence $\lambda_n F_n^{\mathbb{R}} \to 0$ in probability under the conditional law, and bounded convergence yields $\mathbb{E}[\exp(2\pi i \lambda_n F_n^{\mathbb{R}}) \mid S_n = s] \to 1$. The deterministic phase has modulus 1, so $|\mathbb{E}[\,\cdot\,]| \to 1$.

Part (1): subcritical decay (treated in three subcases).

Subcase A: $h = O(\log n)$*.* Theorem 5.1 applied at $\xi = 2^s$ with $A > 1/2 + L^2/4$ yields $|\mathbb{E}[\mathbf{e}_{3^{n+h}}(2^s F_n) \mid S_n = s]| \ll_{A',K,L} n^{-A'}$ for any $A' > 0$.

Subcase B: $h \to \infty$ *and* $\Delta_n = o(n/\log n)$*.* This is Theorem 8.3.

Subcase C: the remaining case, via suffix descent. If Δ_n is not $o(n/\log n)$, then along every subsequence there is a further subsequence with $\Delta_n \ge c\,n/\log n$ for some $c > 0$. On that subsubsequence, $h \to \infty$ (since $\Delta_n \to +\infty$ and $h = h_* n - \Delta_n + O(\sqrt{n\log n})$). Lemma 8.4 applies with $\Delta_n \ge c\,n/\log n$. By Lemma 8.4, choose $R_n = \lfloor h^{3/4}/(\log h)^{3/4} \rfloor$ and $L = \lfloor (h + R_n)/\beta \rfloor$, $J = m - L$, where $\beta = \log_3 2 - 1/2$. Under the bridge law on $\Omega_{m,r}$, the suffix sum $y := \varepsilon_{J+1} + \cdots + \varepsilon_m$ is hypergeometric $\mathrm{Hyper}(m, r, L)$, and Hoeffding gives $\mathbb{P}(y \in \mathcal{T}) = 1 - O(n^{-2})$ for $\mathcal{T} := \{y : |y - L/2| \le C\sqrt{L\log L}\}$.

By Lemma 8.5, $X_m = X_J + 3^{r-y}2^{-J}X_L(\omega)$, hence $P_m = 2^L P_J + 3^{r-y}P_L(\omega)$. Therefore

$$\mathbb{E}[\mathbf{e}_{3^{n+h}}(2P_m) \mid \text{prefix}, y] = \mathbf{e}_{3^{n+h}}(2^{L+1}P_J)\,\mathbb{E}_{\Omega_{L,y}}\mathbf{e}_{3^{y+h+1}}(2P_L),$$

since reducing $r - y \ge 0$ leaves $P_L(\omega) \bmod 3^{N'}$ with $N' := y + h + 1 \le N$. The inner expectation is the resonant smaller-bridge transform up to a phase of modulus 1.

The smaller-bridge entropy gap is $\Delta' := (L+1)\log_3 2 - (y+1) - h = \beta L + (L/2 - y) - h + O(1)$. On $\mathcal{T}$, $|L/2 - y| \le C\sqrt{L\log L} \ll R_n$ by Lemma 8.4, hence $\Delta' = R_n + O(\sqrt{L\log L}) \to +\infty$ and $\Delta' = o(L/\log L)$. Also $s' = L + 1 = 2(y+1) + O(\sqrt{L\log L}) = 2n' + O(\sqrt{n'\log n'})$ with $n' = y + 1$, so s' is in the central window of the smaller bridge.

Theorem 8.3 applies to the smaller bridge: uniformly for $y \in \mathcal{T}$, $\mathbb{E}_{\Omega_{L,y}}\mathbf{e}_{3^{y+h+1}}(2P_L) \to 0$. Off $\mathcal{T}$, the inner expectation has modulus ≤ 1 on an event of probability $O(n^{-2})$. Averaging over y, $|\mathbb{E}[\mathbf{e}_{3^{n+h}}(2P_m) \mid S_n = s]| \le \sup_{y\in\mathcal{T}} |\mathbb{E}_{\Omega_{L,y}}\mathbf{e}_{3^{y+h+1}}(2P_L)| + O(n^{-2}) \to 0$.

The three subcases together cover every $\Delta_n \to +\infty$: Subcase A handles $h = O(\log n)$ (regardless of Δ_n); Subcase B handles $h \to \infty$ with $\Delta_n = o(n/\log n)$; Subcase C handles the remaining case $\Delta_n \ne o(n/\log n)$ by a subsequence argument. Multiplying through by the phase $\mathbf{e}_{3^{n+h}}(3^{n-1})$ completes the proof of part (1). □

8.3. **Implications for the affine character.** By the splitting (4), under $S_n = s$ the multiplicative factor $\mathbf{e}_{3^k}(\xi 2^{-s}x)$ has modulus 1 and is deterministic. Therefore the affine character coefficient $\mathbb{E}[\mathbf{e}_{3^{n+k}}(\xi \mathrm{Aff}_{a_1,\dots,a_n}(x)) \mid S_n = s]$ equals the offset coefficient times this phase, and Theorem 8.6 gives the sharp affine phase transition asserted in Theorem 1.2.

The matching obstruction in part (2) at the entropy line $h = h_* n$ shows that, regardless of any improvement to Tao's renewal estimate, one cannot hope to extend uniform hard-frequency mixing past $h \sim h_* n$.

9. Reduction of all hard frequencies and subexponential decay

The Bernoulli-bridge formula of Theorem 6.2 applies to every primitive $\eta \in (\mathbb{Z}/3^{n+h}\mathbb{Z})^\times$, not only the resonant $\eta = 2^s$. This reduces the entire hard-frequency problem to the analysis of the *primitive ternary bridge transform*

$$\mathcal{C}_{m,r,N}(q) := \mathbb{E}_{\Omega_{m,r}} \mathbf{e}_{3^N}(q\, P_m), \qquad 3 \nmid q. \tag{40}$$

Lemma 6.3 shows that for $t = q\, 2^{m+1}/3^N$ (the "effective real frequency" of $\mathcal{C}_{m,r,N}(2q)$), the multiscale dyadic distance $\mathfrak{D}_m(t) \geq \frac{1}{2} 3^\Delta$ where $\Delta = s \log_3 2 - N$.

We now prove an $o(m/\log m)$-frequency range, which already covers a substantial portion of the hard range. The following lemma is the Bernoulli-bridge analogue of Lemma 8.1; the proof is identical (Stirling expansion of the hypergeometric ratio $\binom{m-R}{r-u}/\binom{m}{r}$ followed by a Hellinger tilt-to-untilt comparison with $p = r/m \approx 1/2$).

Lemma 9.1 (Bernoulli bridge long-block equivalence)**.** *Let $r = m/2 + O(\sqrt{m \log m})$ and $R = R_m$ with $1 \leq R \leq m/2$ and $R \log m/m \to 0$. Under the uniform law on $\Omega_{m,r}$,*

$$\left\| \mathcal{L}((\varepsilon_1, \ldots, \varepsilon_R) \mid \Omega_{m,r}) - \mathrm{Bern}(1/2)^{\otimes R} \right\|_{\mathrm{TV}} \ll \sqrt{\frac{R \log m}{m}} + \frac{R \log m + \log^2 m}{m} + m^{-A}$$

for every fixed $A > 0$.

Theorem 9.2 ($o(m/\log m)$-frequency decay)**.** *Let $r = m/2 + O(\sqrt{m \log m})$, and let $t_m = q_m\, 2^{m+1}/3^{\mathfrak{N}_m}$ with $3 \nmid q_m$. If $|t_m| \to \infty$ and $\log |t_m| = o(m/\log m)$, then $\mathcal{C}_{m,r,\mathfrak{N}_m}(2q_m) \to 0$.*

Proof. Since $P_m = 2^m X_m$, $\mathbf{e}_{3^{\mathfrak{N}_m}}(2q_m P_m) = \exp(2\pi i\, 2q_m P_m/3^{\mathfrak{N}_m}) = \exp(2\pi i t_m X_m)$ where $t_m = q_m \cdot 2^{m+1}/3^{\mathfrak{N}_m}$. The structure of the proof mirrors Theorem 8.3 with the resonant $q = 1$ replaced by an arbitrary primitive frequency q_m.

Step 1: Truncation length. Choose $R = R_m$ satisfying

$$\log |t_m| + \log m \ll R \ll m/\log m, \tag{41}$$

which is feasible by the assumption $\log |t_m| = o(m/\log m)$.

Step 2: Ensemble equivalence. By Lemma 9.1, since $R \ll m/\log m$,

$$\left\| \mathcal{L}((\varepsilon_1, \ldots, \varepsilon_R) \mid \Omega_{m,r}) - \mathrm{Bern}(1/2)^{\otimes R} \right\|_{\mathrm{TV}} \to 0. \tag{42}$$

Step 3: L^α truncation error. Define $X_R := \sum_{j=1}^R \varepsilon_j\, 3^{R_{j-1}}\, 2^{-j}$ and $T_R := X_m - X_R \geq 0$. By Lemma 8.2, applied directly under the bridge law $\Omega_{m,r}$, $\mathbb{E}_{\Omega_{m,r}}[T_R^\alpha] \leq C\rho^R$ for the absolute constants $\alpha \in (0,1)$ and $\rho \in (0,1)$ of that lemma.

Hence

$$\left| \mathbb{E}_{\Omega_{m,r}} e^{2\pi i t_m X_m} - \mathbb{E}_{\Omega_{m,r}} e^{2\pi i t_m X_R} \right| \leq C_\alpha |t_m|^\alpha\, \mathbb{E}_{\Omega_{m,r}}[T_R^\alpha] \leq C_\alpha |t_m|^\alpha \rho^R. \tag{43}$$

Choosing R so that $R \log(1/\rho) \gg \alpha \log |t_m| + \log m$ (consistent with $R \ll m/\log m$ since $\log |t_m| = o(m/\log m)$), the right-hand side is $o(1)$, giving

$$\mathbb{E}_{\Omega_{m,r}} \exp(2\pi i t_m X_m) = \mathbb{E}_{\Omega_{m,r}} \exp(2\pi i t_m X_R) + o(1). \tag{44}$$

Step 4: Reduction to perpetuity. By (42),

$$\mathbb{E}_{\Omega_{m,r}} \exp(2\pi i t_m X_R) = \mathbb{E}_{\mathrm{iid}} \exp(2\pi i t_m \mathcal{F}_R) + o(1), \tag{45}$$

where $\mathcal{F}_R := \sum_{j=1}^R \varepsilon_j\, 3^{R_{j-1}}\, 2^{-j}$ in the i.i.d. Bernoulli(1/2) model. Applying the same L^α estimate to $|\mathcal{F}_\infty - \mathcal{F}_R|$ (directly under the i.i.d. law), and the moving-frequency bound $|\mathbb{E}_{\mathrm{iid}}[e^{2\pi i t_m \mathcal{F}_R} - e^{2\pi i t_m \mathcal{F}_\infty}]| \leq 6\pi |t_m|^\alpha\, \mathbb{E}_{\mathrm{iid}}|\mathcal{F}_\infty - \mathcal{F}_R|^\alpha \ll |t_m|^\alpha m_\alpha^R$, the same choice of R gives $\mathbb{E}_{\mathrm{iid}} \exp(2\pi i t_m \mathcal{F}_R) = \widehat{\mu}_{\mathcal{F}}(t_m) + o(1)$. By Theorem 7.2, $\widehat{\mu}_{\mathcal{F}}(t_m) \to 0$ since $|t_m| \to \infty$. Combining (44) and (45) yields $\mathbb{E}_{\Omega_{m,r}} \exp(2\pi i t_m X_m) \to 0$. $\square$

Remark 9.3 (Range of Theorem 9.2)**.** For Collatz hard frequencies, $t_m = q\,2^{m+1}/3^N$ with $3 \nmid q$ and $N = n+h$. Under $\Delta_n \to +\infty$ we have $\log|t_m| = \log|q| + (m+1)\log 2 - N\log 3 = \log|q| + \Delta_n \log 3 + O(1)$. Hence Theorem 9.2 covers all q with $\log|q| = o(m/\log m)$. The remaining range $\log|q| = \Omega(m/\log m)$ is left to Conjecture 11.1 below.

Corollary 9.4 (Hard-range decay below the entropy line, conditional)**.** *Assume Conjecture 11.1. For $s = 2n + O(\sqrt{n\log n})$ and $h \geq 1$ with $\Delta_n \to +\infty$,*

$$\sup_{\substack{\xi\in\mathbb{Z}/3^{n+k}\mathbb{Z}\\ v_3(\xi)<k}} \left|\mathbb{E}[\mathbf{e}_{3^{n+k}}(\xi F_n) \mid S_n = s]\right| \to 0,$$

and the analogous statement holds for the affine character. The transition at $\Delta_n = 0$ is sharp by Theorem 8.6(2).

10. Density-one and fiberwise flattening

In this section we attack the primitive ternary bridge transform $\mathcal{C}_{m,r,N}(q)$ at exponential frequency $|q|$ where Theorem 9.2 does not apply. The averaging argument we develop does not yet establish the supremum form of HFBD (Conjecture 11.1), but it shows that the bad set of q is exponentially sparse, and remains exponentially sparse inside every 3-adic cylinder. Combined with the suffix self-similarity, this forces any genuine bad sequence q_m to live on a positive-codimension 3-adic exceptional tree.

For this section, we write

$$\mathcal{C}_{L,y,d}(q) := \mathbb{E}_{\Omega_{L,y}} \mathbf{e}_{3^d}(qP_L), \qquad 3 \nmid q,$$

for the local primitive ternary bridge transform on blocks of length L with y ones and ternary modulus 3^d.

10.1. Pair-switch factorization.

Fix $L = 2P$ (the case $L = 2P+1$ adds a single inert bit and is handled identically). Group the bits of $\varepsilon \in \Omega_{L,y}$ into pairs $(\varepsilon_{2p-1}, \varepsilon_{2p})$ for $p = 1,\dots,P$, and let $\sigma_p := \varepsilon_{2p-1} + \varepsilon_{2p} \in \{0,1,2\}$, $C_{p-1} := \sigma_1 + \cdots + \sigma_{p-1}$. Conditional on $\sigma = (\sigma_p)$, the pairs with $\sigma_p = 1$ each carry an independent uniform binary choice between $(1,0)$ and $(0,1)$; pairs with $\sigma_p \in \{0,2\}$ are deterministic.

Lemma 10.1 (Pair-switch valuation)**.** *For any p with $\sigma_p = 1$, switching $(\varepsilon_{2p-1}, \varepsilon_{2p})$ between $(1,0)$ and $(0,1)$ changes $P_L = 2^L X_L$ by exactly $\pm 3^{C_{p-1}} \cdot 2^{L-2p}$, while leaving $C_{p'-1}$ unchanged for all $p' \neq p$.*

Proof. The configuration $(1,0)$ contributes $3^{C_{p-1}} \cdot 2^{L-(2p-1)} = 3^{C_{p-1}} \cdot 2^{L-2p+1}$ to P_L at index $j = 2p-1$ and zero at $j = 2p$; the configuration $(0,1)$ contributes zero at $j = 2p-1$ and $3^{C_{p-1}} \cdot 2^{L-2p}$ at $j = 2p$. Their difference is $3^{C_{p-1}} \cdot 2^{L-2p}$. For any $p' \neq p$, the cumulative count R_{j-1} at $j \in \{2p'-1, 2p'\}$ depends only on $\sigma_1,\dots,\sigma_{p'-1}$, and switching pair p preserves $\sigma_p = 1$. □

Averaging over the binary switches with σ fixed gives

$$\left|\mathbb{E}[\mathbf{e}_{3^d}(qP_L) \mid \sigma]\right|^2 = \prod_{p:\sigma_p=1} \cos^2\!\Big(\pi\,\frac{q\,3^{C_{p-1}}\,2^{L-2p}}{3^d}\Big). \tag{46}$$

Let $A(\sigma) := \#\{p : \sigma_p = 1\}$ denote the number of active pairs.

10.2. Ternary digit averaging.

The averaging identity

$$\frac{1}{3}\sum_{a=0}^{2} \cos^2\!\big(\pi(x + a/3)\big) = \frac{1}{2} \tag{47}$$

follows from $\cos^2\theta = (1+\cos 2\theta)/2$ and $\sum_{a=0}^{2} e^{2\pi i a/3} = 0$.

Lemma 10.2 (Cylinder averaging)**.** *Fix* $0 \le d_0 < d$, $D := d - d_0$, *and a residue* $q_0 \in \mathbb{Z}/3^{d_0}\mathbb{Z}$ *with* $\gcd(q_0, 3) = 1$. *For any pair-sum pattern* σ,

$$\frac{1}{\#\{q \bmod 3^d : q \equiv q_0 \pmod{3^{d_0}}\}} \sum_{\substack{q \bmod 3^d \\ q \equiv q_0 \pmod{3^{d_0}}}} \prod_{p:\sigma_p=1} \cos^2\!\Big(\pi \frac{q\, 3^{C_{p-1}}\, 2^{L-2p}}{3^d}\Big) \le 2^{-A_D(\sigma)}, \tag{48}$$

where $A_D(\sigma) := \#\{p : \sigma_p = 1,\ C_{p-1} < D\}$.

Proof. Write $\mathrm{Active}(\sigma) := \{p : \sigma_p = 1\}$ and $A = A(\sigma) := |\mathrm{Active}(\sigma)|$. For $p \in \mathrm{Active}(\sigma)$ set $\alpha_p := 3^{C_{p-1}} \cdot 2^{L-2p} \in \mathbb{Z}$. Note $\gcd(2, 3) = 1$, so the 3-adic valuation of α_p is exactly C_{p-1}. Since each active pair increments the cumulative count by 1, the values $\{C_{p-1} : p \in \mathrm{Active}(\sigma)\}$ are pairwise distinct.

Use the identity $\cos^2\theta = (1 + \cos 2\theta)/2$ at $\theta = \pi q\alpha_p/3^d$ and expand the product:

$$\begin{aligned}\prod_{p\in\mathrm{Active}(\sigma)} \cos^2\!\Big(\frac{\pi q\alpha_p}{3^d}\Big) &= \frac{1}{2^A} \sum_{S\subseteq\mathrm{Active}(\sigma)} \prod_{p\in S} \cos\!\Big(\frac{2\pi q\alpha_p}{3^d}\Big) \\ &= \frac{1}{2^A} \sum_{S\subseteq\mathrm{Active}(\sigma)} \frac{1}{2^{|S|}} \sum_{\varepsilon\in\{\pm1\}^S} \mathbf{e}_{3^d}\big(q\, K_S(\varepsilon)\big),\end{aligned}$$

where $K_S(\varepsilon) := \sum_{p\in S} \varepsilon_p\alpha_p \in \mathbb{Z}$.

For the cylinder average we decompose $q = q_0 + 3^{d_0} q'$ with $q' \in \mathbb{Z}/3^D\mathbb{Z}$ uniformly distributed, and use orthogonality of additive characters on $\mathbb{Z}/3^D\mathbb{Z}$:

$$\mathbb{E}_{q\equiv q_0 \pmod{3^{d_0}}} \mathbf{e}_{3^d}(qK_S) = \mathbf{e}_{3^d}(q_0K_S) \cdot \mathbf{1}\big[3^D \mid K_S\big]. \tag{49}$$

We claim that $3^D \mid K_S(\varepsilon)$ if and only if every $p \in S$ satisfies $C_{p-1} \ge D$. Sufficiency is immediate since each α_p is then divisible by 3^D. For necessity, suppose some $p \in S$ has $C_{p-1} < D$. Let p_* be the element of S with smallest C_{p-1}. Then $v_3(\alpha_{p_*}) = C_{p_*-1}$ and every other $p \in S$ has $v_3(\alpha_p) > C_{p_*-1}$ since the C_{p-1}-values are distinct. Hence $v_3(K_S(\varepsilon)) = v_3(\varepsilon_{p_*}\alpha_{p_*}) = C_{p_*-1} < D$, contradicting $3^D \mid K_S(\varepsilon)$.

Therefore the only surviving subsets S in the expansion are $S \subseteq \{p \in \mathrm{Active}(\sigma) : C_{p-1} \ge D\}$, of which there are $2^{A-A_D(\sigma)}$. For each such S and each ε, the character $\mathbf{e}_{3^d}(q_0K_S)$ has modulus 1. Combining,

$$\Big|\mathbb{E}_{q\equiv q_0} \prod_{p\in\mathrm{Active}(\sigma)} \cos^2\!\Big(\frac{\pi q\alpha_p}{3^d}\Big)\Big| \le \frac{1}{2^A} \sum_{S\subseteq\{C_{p-1}\ge D\}} \frac{1}{2^{|S|}} \cdot 2^{|S|} = \frac{2^{A-A_D(\sigma)}}{2^A} = 2^{-A_D(\sigma)},$$

proving (48). The constraint $\gcd(q_0, 3) = 1$ is needed only to ensure that the cylinder $q \equiv q_0 \pmod{3^{d_0}}$ contains primitive elements; the orthogonality argument itself does not use it. □

10.3. **Bridge generating function.** Recall the conditional law of σ given $\sum_p \sigma_p = y$ is

$$\mathbb{P}_{\Omega_{L,y}}(\sigma) = \frac{2^{A(\sigma)}}{[z^y](1 + 2z + z^2)^P} \cdot \mathbf{1}\Big[\sum_p \sigma_p = y\Big], \tag{50}$$

since each pair contributes a factor $(1 + 2z + z^2) = (1 + z)^2$ to $(1 + z)^L = \sum_y \#\Omega_{L,y}\, z^y$, with the $2z$ term coming from the two configurations of an active pair.

Theorem 10.3 (Density-one HFBD; averaged flattening)**.** *Let* $L = 2P$, $y = P + O(\sqrt{P \log P})$, *and* $d > y$. *There exists* $c > 0$ *such that*

$$\frac{1}{\varphi(3^d)} \sum_{\substack{q \bmod 3^d \\ 3\nmid q}} |\mathcal{C}_{L,y,d}(q)|^2 \ll L^C\, e^{-cL}. \tag{51}$$

Consequently, for every threshold $\tau > 0$,

$$\#\{q \bmod 3^d : 3 \nmid q,\ |\mathcal{C}_{L,y,d}(q)| > \tau\} \ll \tau^{-2} L^C\, e^{-cL}\, \varphi(3^d).$$

Proof. By Cauchy–Schwarz applied to $\mathbb{E}_{\Omega_{L,y}}\mathbb{E}_\sigma[\,\cdot \mid \sigma]$,

$$|\mathcal{C}_{L,y,d}(q)|^2 \le \mathbb{E}_\sigma\big|\mathbb{E}[\mathbf{e}_{3^d}(qP_L) \mid \sigma]\big|^2.$$

Lemma 10.2 with $d_0 = 0$ gives an upper bound on the average over *all* residues modulo 3^d (not only primitive ones). Since the summands are nonnegative and primitive residues have density $\varphi(3^d)/3^d = 2/3$,

$$\frac{1}{\varphi(3^d)} \sum_{\substack{q \bmod 3^d \\ 3\nmid q}} |\mathcal{C}_{L,y,d}(q)|^2 \le \frac{3^d}{\varphi(3^d)} \cdot \frac{1}{3^d} \sum_{q \bmod 3^d} |\mathcal{C}_{L,y,d}(q)|^2 = \frac{3}{2} \cdot \frac{1}{3^d} \sum_{q \bmod 3^d} |\mathcal{C}_{L,y,d}(q)|^2.$$

Applying Lemma 10.2 with $d_0 = 0$ to the right-hand side, and using $D = d > y \ge A(\sigma)$ so $A_D(\sigma) = A(\sigma)$,

$$\frac{1}{\varphi(3^d)} \sum_{3\nmid q} |\mathcal{C}_{L,y,d}(q)|^2 \le \frac{3}{2}\, \mathbb{E}_\sigma\big[2^{-A(\sigma)}\big].$$

By (50), the conditional law of σ on $\sum_p \sigma_p = y$ assigns weight proportional to $2^{A(\sigma)}$. Therefore

$$\mathbb{E}_\sigma\big[2^{-A(\sigma)}\big] = \frac{\sum_\sigma 2^{-A(\sigma)} \cdot 2^{A(\sigma)} \mathbf{1}[\sum \sigma_p = y]}{\sum_\sigma 2^{A(\sigma)} \mathbf{1}[\sum \sigma_p = y]} = \frac{[z^y](1+z+z^2)^P}{[z^y](1+2z+z^2)^P}, \tag{52}$$

since the unweighted single-pair generating function (counting each $\sigma_p \in \{0,1,2\}$ with multiplicity 1) is $1+z+z^2$, and the weighted single-pair generating function counting bridges with the 2-fold internal degree of freedom for active pairs is $(1+2z+z^2) = (1+z)^2$.

Lower bound on the denominator. $[z^y](1+z)^{2P} = \binom{2P}{y}$. For $y = P + \tilde\sigma$ with $\tilde\sigma = O(\sqrt{P\log P})$, Stirling gives

$$\binom{2P}{P+\tilde\sigma} = \frac{4^P}{\sqrt{\pi P}}\, e^{-\tilde\sigma^2/P}\big(1 + O(P^{-1/2})\big) \gg \frac{4^P}{\sqrt{P}} \cdot e^{-O(\log P)}. \tag{53}$$

Upper bound on the numerator. The trinomial coefficient $a_P(y) := [z^y](1+z+z^2)^P$ is the probability generating function for the sum of P i.i.d. uniform variables on $\{0,1,2\}$, scaled by 3^P. That sum has mean P and variance $2P/3$. By the local central limit theorem on the lattice $\mathbb{Z}$ (see e.g. [6, Theorem 2.5.2]),

$$a_P(y) = \frac{3^P}{\sqrt{2\pi\cdot 2P/3}}\, e^{-3(y-P)^2/(4P)}\big(1 + O(P^{-1/2})\big) \ll \frac{3^P}{\sqrt{P}}. \tag{54}$$

Combining. Dividing (54) by (53),

$$\mathbb{E}_\sigma\big[2^{-A(\sigma)}\big] \ll \frac{3^P/\sqrt{P}}{4^P/\sqrt{P}} \cdot e^{O(\log P)} = \Big(\frac{3}{4}\Big)^P \cdot P^C = e^{-(L/2)\log(4/3)} \cdot L^C,$$

which proves (51) with any $c < \frac{1}{2}\log(4/3) \approx 0.144$. The Markov inequality $\#\{q : |\mathcal{C}| > \tau\} \le \tau^{-2}\sum |\mathcal{C}|^2$ gives the second statement. □

10.4. **Fiberwise flattening.**

Theorem 10.4 (Fiberwise flattening)**.** *Let $L = 2P$, $y = P + O(\sqrt{P\log P})$, $0 \le d_0 < d$, $D := d - d_0$. For any primitive $q_0 \in (\mathbb{Z}/3^{d_0}\mathbb{Z})^\times$,*

$$\frac{1}{\#\{q \bmod 3^d : q \equiv q_0 \pmod{3^{d_0}}\}} \sum_{\substack{q \bmod 3^d \\ q \equiv q_0 \pmod{3^{d_0}}}} |\mathcal{C}_{L,y,d}(q)|^2 \ll L^C\, e^{-c\min(D,L)}. \tag{55}$$

Proof. By Cauchy–Schwarz and Lemma 10.2,

$$\text{LHS of (55)} \le \mathbb{E}_\sigma\big[2^{-A_D(\sigma)}\big].$$

Reduction to the leading M pairs. Set $M := \lfloor \min(D,P)/2 \rfloor$. For $p \le M$ the cumulative count satisfies $C_{p-1} = \sigma_1 + \cdots + \sigma_{p-1} \le 2(p-1) \le 2M - 2 < 2M \le D$, so every active pair among the first M pairs lies inside the set $\{C_{p-1} < D\}$ and hence contributes to $A_D(\sigma)$. Writing $B_M(\sigma) := \#\{p \le M : \sigma_p = 1\}$ we obtain $A_D(\sigma) \ge B_M(\sigma)$, hence

$$\mathbb{E}_\sigma\big[2^{-A_D(\sigma)}\big] \le \mathbb{E}_\sigma\big[2^{-B_M(\sigma)}\big]. \tag{56}$$

Generating function for $\mathbb{E}_\sigma 2^{-B_M(\sigma)}$. Using (50) and the fact that pairs are independent given their sums,

$$\mathbb{E}_\sigma\big[2^{-B_M(\sigma)}\big] = \frac{[z^y]\big[(1+z+z^2)^M(1+2z+z^2)^{P-M}\big]}{[z^y](1+2z+z^2)^P}. \tag{57}$$

The first M pair factors are $(1+2z+z^2)$ in the denominator and $(1+z+z^2)$ in the numerator (because the $\sigma = 1$ contribution $2z$ becomes z after reweighting by 2^{-1}); the remaining $P-M$ pair factors are $(1+2z+z^2)$ on both sides.

Saddle-point estimate. By Lemma A.1 (Appendix A),

$$[z^y](1+z+z^2)^M(1+2z+z^2)^{P-M} \ll P^C \cdot \frac{3^M \cdot 4^{P-M}}{\sqrt{P}}.$$

Dividing by the denominator $\binom{2P}{y} \gg 4^P/(P^C\sqrt{P})$ (Stirling, central window),

$$\mathbb{E}_\sigma\big[2^{-B_M(\sigma)}\big] \ll P^C \cdot \frac{3^M \cdot 4^{P-M}}{4^P} = P^C\Big(\frac{3}{4}\Big)^M \ll L^C\, e^{-c\min(D,L)}, \tag{58}$$

since $M = \Theta(\min(D,L))$ and $(3/4)^M = e^{-M\log(4/3)}$. This proves (55). □

10.5. **Multiscale pruning of exceptional sets.** The fiberwise estimate of Theorem 10.4 is closed under composition and yields a multiscale codimension bound for the bad set.

Corollary 10.5 (Exceptional-tree codimension)**.** *Fix a chain of depths* $0 = d_0 < d_1 < \cdots < d_T$ *and bridge blocks* (L_j, y_j) *with* $y_j = L_j/2 + O(\sqrt{L_j \log L_j})$*. Set* $H_j := \min(d_j - d_{j-1}, L_j)$ *and* $\tau_j := e^{-cH_j/4}$*. Define*

$$\mathcal{E}_T := \big\{q \in (\mathbb{Z}/3^{d_T}\mathbb{Z})^\times : |\mathcal{C}_{L_j,y_j,d_j}(q \bmod 3^{d_j})| > \tau_j \textit{ for every } 1 \le j \le T\big\}.$$

If $H_j \gg \log L_j$ *for every* j*, then*

$$\#\mathcal{E}_T \ll \varphi(3^{d_T})\, \exp\left(-c' \sum_{j=1}^T H_j\right) \tag{59}$$

for an absolute $c' > 0$*. In particular, if* $\sum_j H_j \asymp d_T$*, then* $\#\mathcal{E}_T \ll 3^{(1-\kappa)d_T}$ *for some* $\kappa > 0$*.*

Proof. We iterate Theorem 10.4 from $j = 1$ to $j = T$, conditioning at each step on the residue class fixed at the previous level.

Single-level density. At level j, fix $q_{j-1} \in \mathbb{Z}/3^{d_{j-1}}\mathbb{Z}$ with $\gcd(q_{j-1}, 3) = 1$. The number of residues in $\mathbb{Z}/3^{d_j}\mathbb{Z}$ above q_{j-1} is exactly $3^{d_j - d_{j-1}}$. By Theorem 10.4 applied with $(L, y, d_0, d) := (L_j, y_j, d_{j-1}, d_j)$, $D = d_j - d_{j-1}$, the average of $|\mathcal{C}_{L_j,y_j,d_j}(q)|^2$ over this cylinder is $\ll L_j^C\, e^{-cH_j}$ where $H_j = \min(D, L_j)$. Markov's inequality therefore gives

$$\#\big\{q \equiv q_{j-1} \pmod{3^{d_{j-1}}} : |\mathcal{C}_{L_j,y_j,d_j}(q \bmod 3^{d_j})| > \tau_j\big\} \le \tau_j^{-2}\, L_j^C\, e^{-cH_j} \cdot 3^{d_j - d_{j-1}}. \tag{60}$$

Substituting $\tau_j = e^{-cH_j/4}$ and using $H_j \gg \log L_j$ to absorb the polynomial factor,

$$\#\big\{q \equiv q_{j-1} \pmod{3^{d_{j-1}}} : |\mathcal{C}| > \tau_j\big\} \le e^{-c'' H_j} \cdot 3^{d_j - d_{j-1}} \tag{61}$$

for some $c'' > 0$.

Iterating the chain. At level $j = 0$ all $\varphi(3^{d_0})$ primitive residues are eligible. At each subsequent level the bad lifts in each $3^{d_{j-1}}$-cylinder number at most $e^{-c'' H_j} \cdot 3^{d_j - d_{j-1}}$. By induction on j,

$$\begin{aligned}\#\mathcal{E}_T &\le \varphi(3^{d_0}) \prod_{j=1}^{T} \bigl(e^{-c''H_j} \cdot 3^{d_j - d_{j-1}}\bigr) = \varphi(3^{d_0}) \cdot 3^{d_T - d_0} \cdot \prod_{j=1}^{T} e^{-c''H_j} \\ &\le \varphi(3^{d_T}) \cdot \exp\Bigl(-c'' \sum_{j=1}^{T} H_j\Bigr),\end{aligned}$$

using $\varphi(3^{d_0}) \cdot 3^{d_T - d_0} \le \varphi(3^{d_T}) \cdot (3/2)$ and absorbing the constant.

Linear codimension. If $\sum_j H_j \asymp d_T$, then $\varphi(3^{d_T}) \cdot e^{-c' \sum H_j} \ll 3^{d_T} \cdot e^{-c'' d_T} = 3^{d_T(1 - c''/\log 3)} = 3^{(1-\kappa)d_T}$ with $\kappa = c''/\log 3 > 0$. □

Remark 10.6 (Implication for HFBD)**.** Corollary 10.5 shows that any sequence q_m violating HFBD-sup must, in the bridge representation, survive a positive-codimension exceptional tree at all scales, in the sense that $q_m \bmod 3^{d_j}$ avoids the union of bad lifts at every level j. The structure of such surviving residues mirrors Tao's black-triangle chains in [11, §7]: nested 3-adic resonances of arithmetic type. The natural way to close the tree is to show such a sequence must be asymptotically resonant, i.e. $q_m \equiv \pm 2^a \pmod{3^{d_j}}$ for arbitrarily large depth d_j, in which case the resonant analysis of §8 applies.

Remark 10.7 (A structural ceiling for second-moment methods)**.** The codimension $\kappa > 0$ in Corollary 10.5 is bounded by the ratio of the per-step pair-switch contraction to the ternary branching: each unit of 3-adic depth introduces 3 branches against a pair-switch contraction of 1/2 per active pair, giving an asymptotic survival rate of order $(3/2)^{d_T/2}$ for the absolute size of $\mathcal{E}_T$. In particular, second-moment methods of the type used in Theorems 10.3–10.4 yield positive codimension but cannot yield $\#\mathcal{E}_T = O(1)$, which is what HFBD-sup would require. The supremum form of HFBD therefore requires a rigidity input beyond L^2-flattening, and the natural candidate is the resonant-capture statement of Remark 10.6.

11. The remaining open range

The exact suffix self-similarity (Lemma 8.5) reduces the exponential range of $|q|$ to a Diophantine condition. For $L \to \infty$ and $y = L/2 + O(\sqrt{L \log m})$, define $d_L := y + h + 1$ and let $q_{d_L} \in \mathbb{Z}$ be the representative of $2q \pmod{3^{d_L}}$ in $(-3^{d_L}/2, 3^{d_L}/2]$, so $3 \nmid q_{d_L}$.

Conjecture 11.1 (High-frequency primitive ternary bridge decay; HFBD)**.** *Let $r = m/2 + O(\sqrt{m \log m})$ and $3 \nmid q$. If $m \log 2 - N \log 3 \to +\infty$, then*

$$\sup_{\substack{q \in \mathbb{Z}/3^N\mathbb{Z} \\ 3 \nmid q}} |\mathcal{C}_{m,r,N}(q)| \to 0.$$

Theorem 9.2 verifies HFBD in the subexponential range $\log |q| = o(m/\log m)$, and Theorems 10.3–10.4 give a strong density-one form of HFBD in the exponential range as well: the bad set of primitive frequencies is exponentially sparse in every 3-adic cylinder, and surviving frequencies have positive 3-adic codimension (Corollary 10.5). The remaining content of HFBD is therefore a rigidity statement on the multiscale exceptional tree.

Remark 11.2 (Status after Section 10)**.** By Theorem 9.2, HFBD holds for $\log |q| = o(m/\log m)$. By Theorems 10.3 and 10.4, for $\log |q| = \Theta(m)$ the bad set of primitive residues is exponentially sparse in every 3-adic cylinder; iterating the fiberwise estimate over a chain of suffix scales gives Corollary 10.5, which forces any HFBD-violating sequence (q_m) to survive a positive-codimension exceptional tree. The natural completion (Remark 10.6) is a rigidity theorem asserting that any such surviving sequence must be asymptotically captured by the resonant family $\{\pm 2^a\}$, in which case the analysis of Section 8 closes the remaining branch. The Brémont/Li–Sahlsten/Rapaport classification [1, 5, 8] suggests that the obstruction, if any, is of arithmetic Pisot type and hence rigid; the IFS (36) is not in Pisot form, giving a structural reason to expect HFBD to hold.

Implications for Tao's Remark 1.16. A natural-density upgrade of Tao's theorem requires fine-scale mixing of the entire affine map $\mathrm{Aff}_{a_1,\dots,a_n}$. The results of this paper show:

- the multiplicative marginal $2^{-S_n} \pmod{3^k}$ alone cannot provide such mixing past $3^k \sim \sqrt{n}$ (Theorem 1.1);
- under microcanonical conditioning the multiplicative obstruction disappears and the worst (resonant) hard frequency exhibits a sharp phase transition at the entropy line $h_* n$ (Theorem 8.6);
- the full hard-frequency problem reduces to the primitive ternary bridge transform (40) and is unconditionally solved in the subexponential range (Theorem 9.2);
- in the exponential range, the bad set of primitive residues is exponentially sparse in every 3-adic cylinder, so a multiscale exceptional tree is forced to have positive codimension (Theorems 10.3, 10.4, Corollary 10.5);
- the matching obstruction at $h > h_* n$ identifies the sharp limit of any uniform decay, regardless of technical strength of Tao's renewal estimate.

What remains for the supremum form of HFBD is a rigidity statement asserting that any HFBD-violating sequence must be asymptotically captured by the resonant family $\{\pm 2^a\}$. A proof of this rigidity would close hard-frequency affine mixing throughout the entire entropy-allowed range and provide the missing input toward Tao's Remark 1.16.

Appendix A. Technical lemmas

We collect several technical estimates used in the body of the paper.

A.1. **Local central limit theorem for S_n.** The variable $S_n = a_1 + \cdots + a_n$ with a_i i.i.d. Geom(2) is supported on $\{n, n+1, \dots\}$ with $\mathbb{E}S_n = 2n$ and $\mathrm{Var}S_n = 2n$. Its characteristic function is $\phi_n(\theta) = \left(\frac{1}{2}e^{i\theta}/(1-\frac{1}{2}e^{i\theta})\right)^n$, satisfying $|\phi_n(\theta)| \le e^{-cn\sin^2(\theta/2)}$ for $|\theta| \le \pi$. By Fourier inversion and the standard local CLT for lattice distributions (see e.g. [6, Theorem 2.5.2]),

$$\mathbb{P}(S_n = s) = \frac{1}{\sqrt{4\pi n}} \exp\Big(-\frac{(s-2n)^2}{4n}\Big)\left(1 + O(n^{-1/2})\right) \tag{62}$$

uniformly for $|s-2n| \le n^{2/3}$ (say). For $|s-2n| \le L\sqrt{n\log n}$, this gives $\mathbb{P}(S_n = s) \gg n^{-1/2-L^2/4}$, the bound used in the proof of Theorem 5.1.

A.2. **Hoeffding for hypergeometric.** Let $y \sim \mathrm{Hyper}(m, r, L)$ denote the number of ones in a uniform L-subset of $\{0,1\}^m$ subject to total r. Equivalently, y is the number of red balls drawn in L draws without replacement from an urn with r red and $m-r$ white balls. Hoeffding's inequality (cf. [6, Theorem 12.4]) gives

$$\mathbb{P}\big(|y - Lr/m| > t\big) \le 2\exp(-2t^2/L) \quad \text{for all } t > 0. \tag{63}$$

A.3. **L^α moment of the perpetuity tail.** For $A \sim$ Geom(2) on $\{1,2,\dots\}$ and $0 < \alpha < 1$,

$$m_\alpha := \mathbb{E}\big[(3\cdot 2^{-A})^\alpha\big] = \frac{3^\alpha}{2^{\alpha+1}-1} < 1. \tag{64}$$

This follows from $\mathbb{E}[2^{-\alpha A}] = 2^{-(\alpha+1)}/(1-2^{-(\alpha+1)}) = 1/(2^{\alpha+1}-1)$. At $\alpha = 0$ and $\alpha = 1$, $m_\alpha = 1$; convexity of $f(\alpha) := \log m_\alpha = \alpha\log 3 - \log(2^{\alpha+1}-1)$ on $[0,1]$ (a direct second-derivative check shows $f'' > 0$, then $f(0) = f(1) = 0$ implies $f < 0$ on the interior) gives $m_\alpha < 1$ for $\alpha \in (0,1)$.

A.4. **Mixed saddle-point estimate.**

Lemma A.1 (Mixed saddle-point)**.** *Let $G_{P,M}(z) := (1+z+z^2)^M(1+2z+z^2)^{P-M}$ for $0 \le M \le P$. Uniformly for $y = P + O(\sqrt{P\log P})$,*

$$[z^y]G_{P,M}(z) \ll P^C\,\frac{3^M\cdot 4^{P-M}}{\sqrt{P}}$$

for an absolute constant C.

Proof. Write $G_{P,M}(z) = 3^M 4^{P-M}\,\mathbb{E}z^{U_M+V_{P-M}}$, where U_M is the sum of M i.i.d. uniform $\{0,1,2\}$ variables and $V_{P-M} \sim \mathrm{Bin}(2(P-M),1/2)$ is independent. Hence $[z^y]G_{P,M}(z) = 3^M 4^{P-M}\mathbb{P}(U_M + V_{P-M} = y)$, and it suffices to show $\sup_y \mathbb{P}(U_M + V_{P-M} = y) \ll P^{-1/2}$.

If $P - M \geq P/2$, $\sup_y \mathbb{P}(U_M + V_{P-M} = y) \leq \sup_v \mathbb{P}(V_{P-M} = v)$, and Stirling for binomial coefficients gives this $\ll (P-M)^{-1/2} \ll P^{-1/2}$.

If $M \geq P/2$, $\sup_y \mathbb{P}(U_M + V_{P-M} = y) \leq \sup_u \mathbb{P}(U_M = u)$. The characteristic function of a single uniform $\{0,1,2\}$ variable is $\phi(\theta) = (1 + e^{i\theta} + e^{2i\theta})/3 = e^{i\theta}(1 + 2\cos\theta)/3$, so $|\phi(\theta)| = |1 + 2\cos\theta|/3$. There are $c, c' > 0$ with $|\phi(\theta)| \leq e^{-c\theta^2}$ for $|\theta| \leq 1$ and $|\phi(\theta)| \leq 1 - c'$ for $1 \leq |\theta| \leq \pi$. Fourier inversion gives

$$\sup_u \mathbb{P}(U_M = u) \leq \frac{1}{2\pi}\int_{-\pi}^{\pi} |\phi(\theta)|^M d\theta \ll \int_{-1}^{1} e^{-cM\theta^2}\, d\theta + (1-c')^M \ll M^{-1/2} \ll P^{-1/2}.$$

Combining the two cases proves the lemma. □

A.5. **Saddle-point estimate for trinomial coefficients.** For the polynomial $(1+z+z^2)^P$, the coefficient $a_P(y) := [z^y](1+z+z^2)^P$ is the probability generating function of P i.i.d. uniform$\{0,1,2\}$ variables, scaled by 3^P. The local CLT for lattice distributions gives, for $y = P + \tilde{\sigma}$ with $\tilde{\sigma} = O(\sqrt{P\log P})$,

$$a_P(y) = \frac{3^P}{\sqrt{2\pi \cdot 2P/3}}\, \exp\Bigl(-\frac{3\tilde{\sigma}^2}{4P}\Bigr)\bigl(1 + O(P^{-1/2})\bigr). \tag{65}$$

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